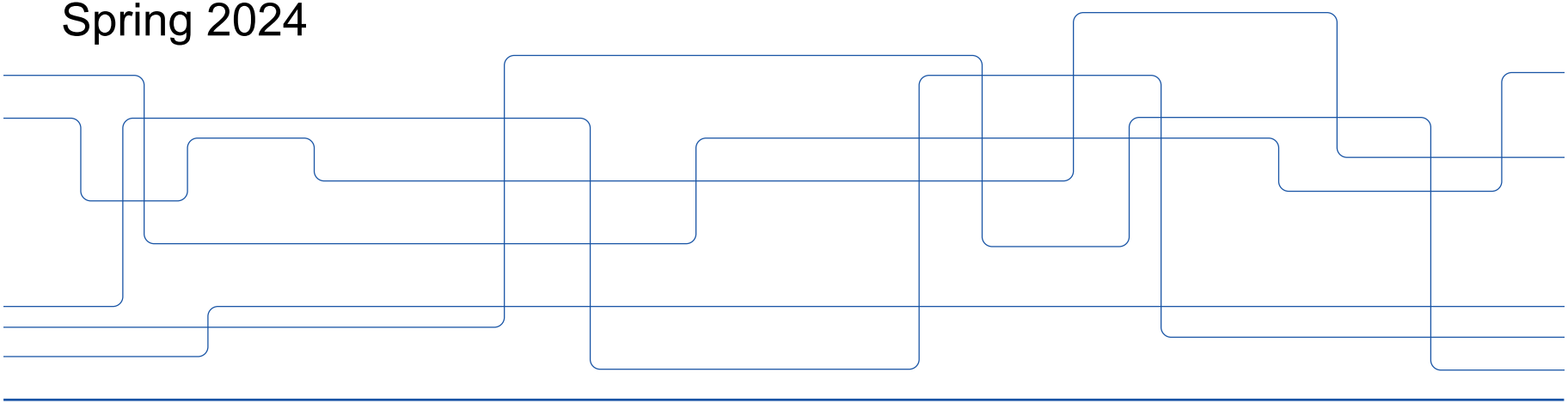
CM2014

Modelling Systems

System dynamics, discrete event simulation, agent-based (network) modelling

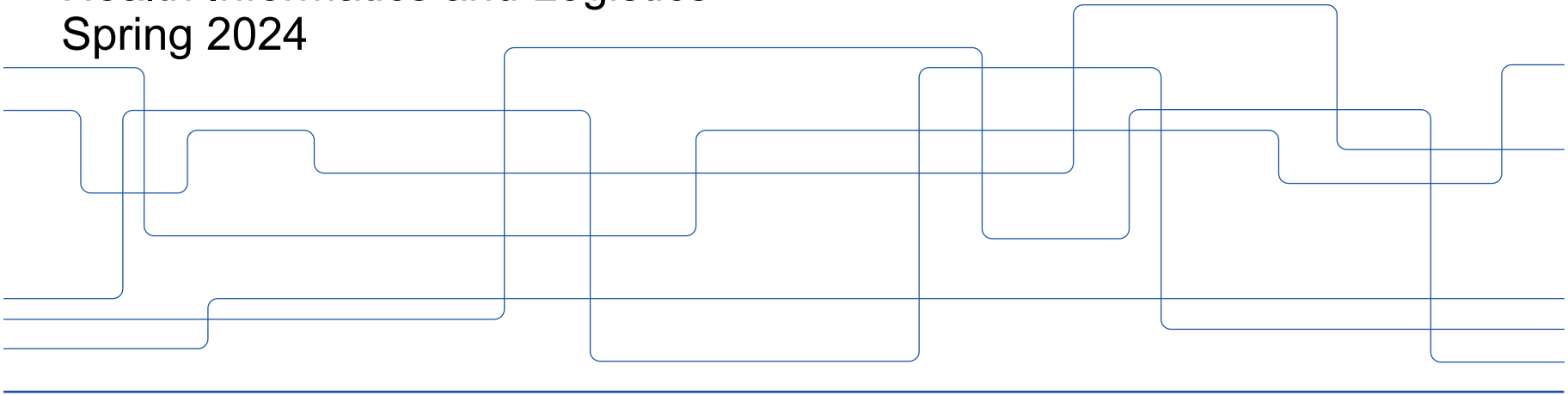
Spring 2024



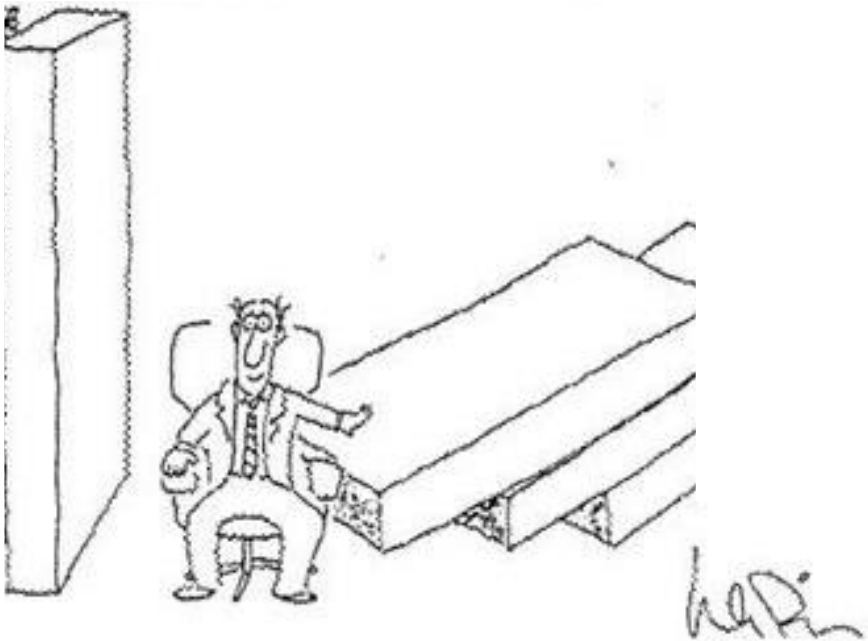
CM2014

System dynamic modelling and simulation
Lecture 1: basic principles and concepts

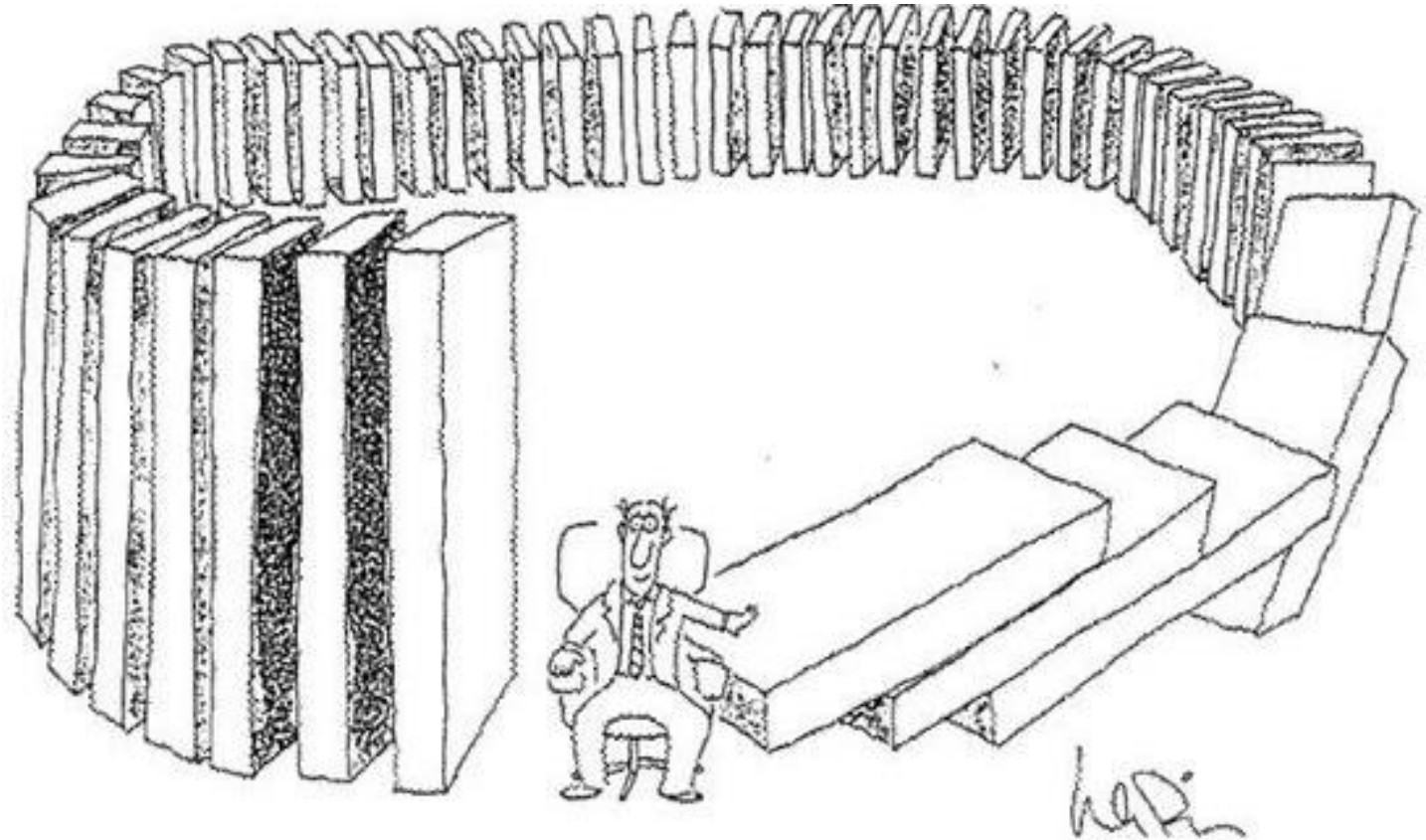
Adam S. Darwich
Associate Professor of Health Systems Engineering
Health Informatics and Logistics
Spring 2024



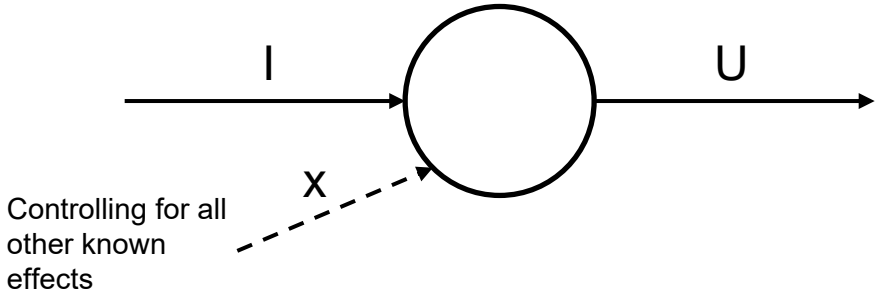
Reductionist approach: stimulation-response



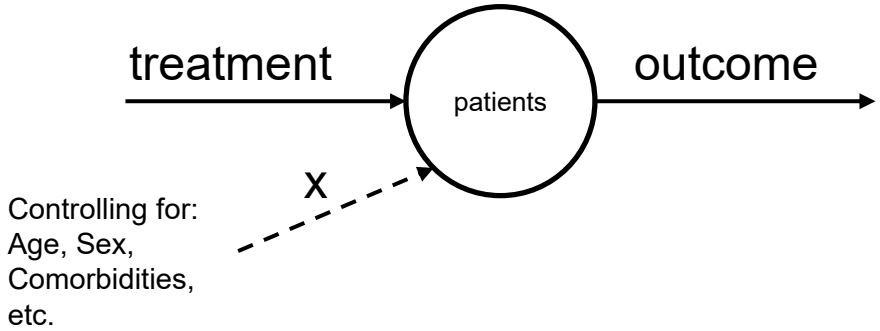
Systems thinking: emergent behaviour from interacting entitites



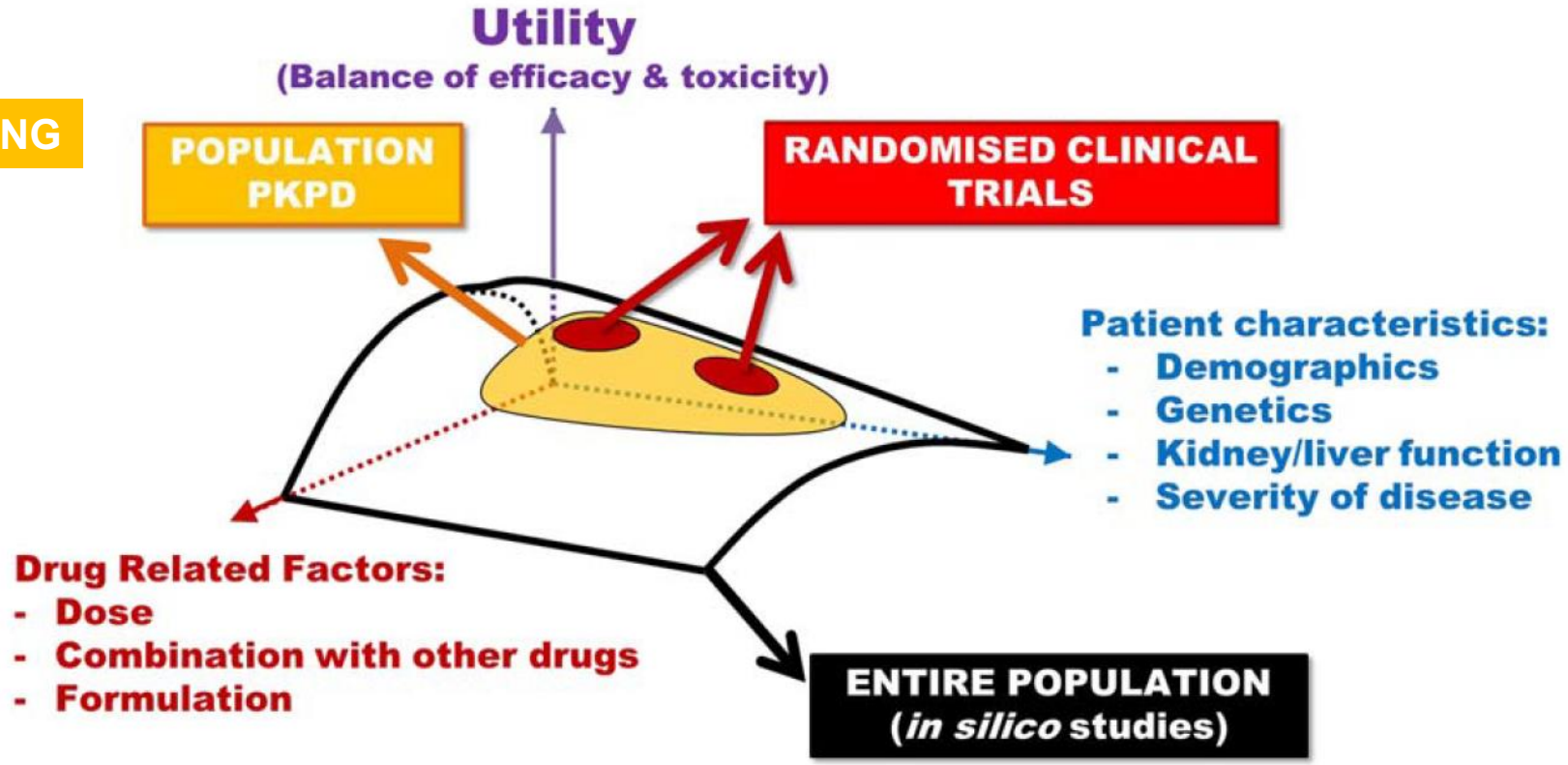
The traditional scientific method



Example:

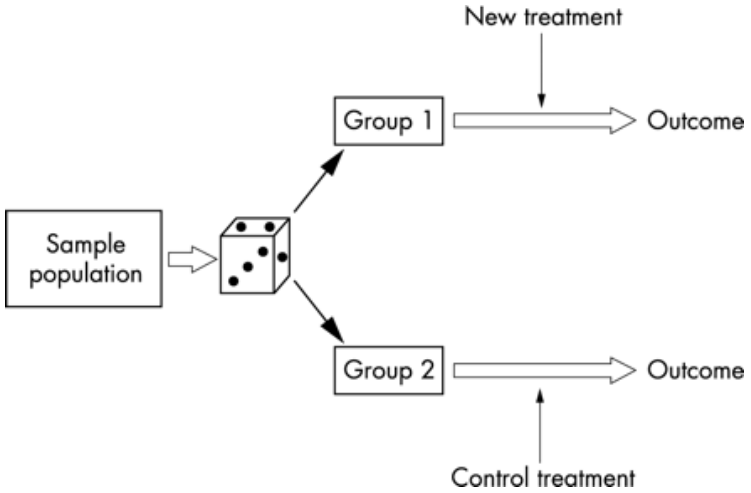


MODELLING



Darwich *et al*. 2017

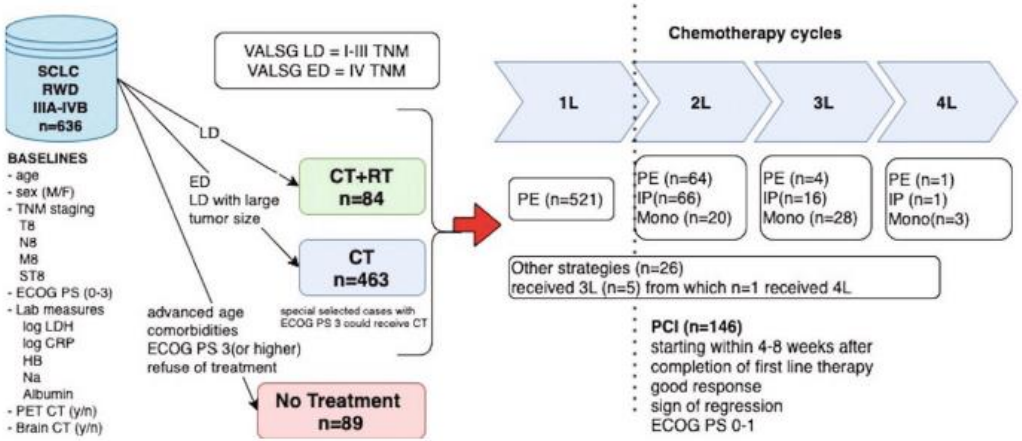
Randomised controlled trials (RCTs) vs. Real-world healthcare data (RWD)



RESEARCH SERIES

Designing a research project: randomised controlled trials and their principles

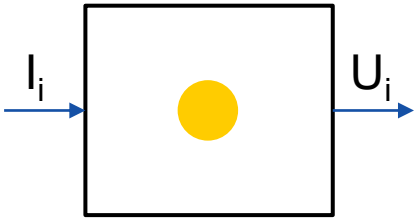
J M Kendall

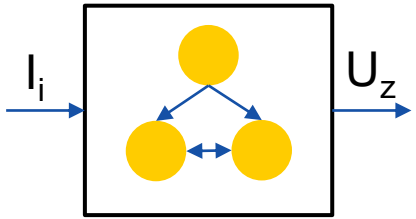


A novel analytical framework for risk stratification of real-world data using machine learning: A small cell lung cancer study

Luca Marzano[1] | Adam S. Darwich[1] | Salomon Tendler[2] | Asaf Dan[2] | Rolf Lewensohn[2] | Luigi De Petris[2] | Jayanth Raghothama[1] | Sebastiaan Meijer[1]

Systems theory





"*We often think that when we have completed our study of one, we know all about two, because 'two' is 'one and one'. We forget that we still have to make a study of 'and'. Secondary physics is the study of "and", that is to say, of organisation.*" – Arthur Eddington



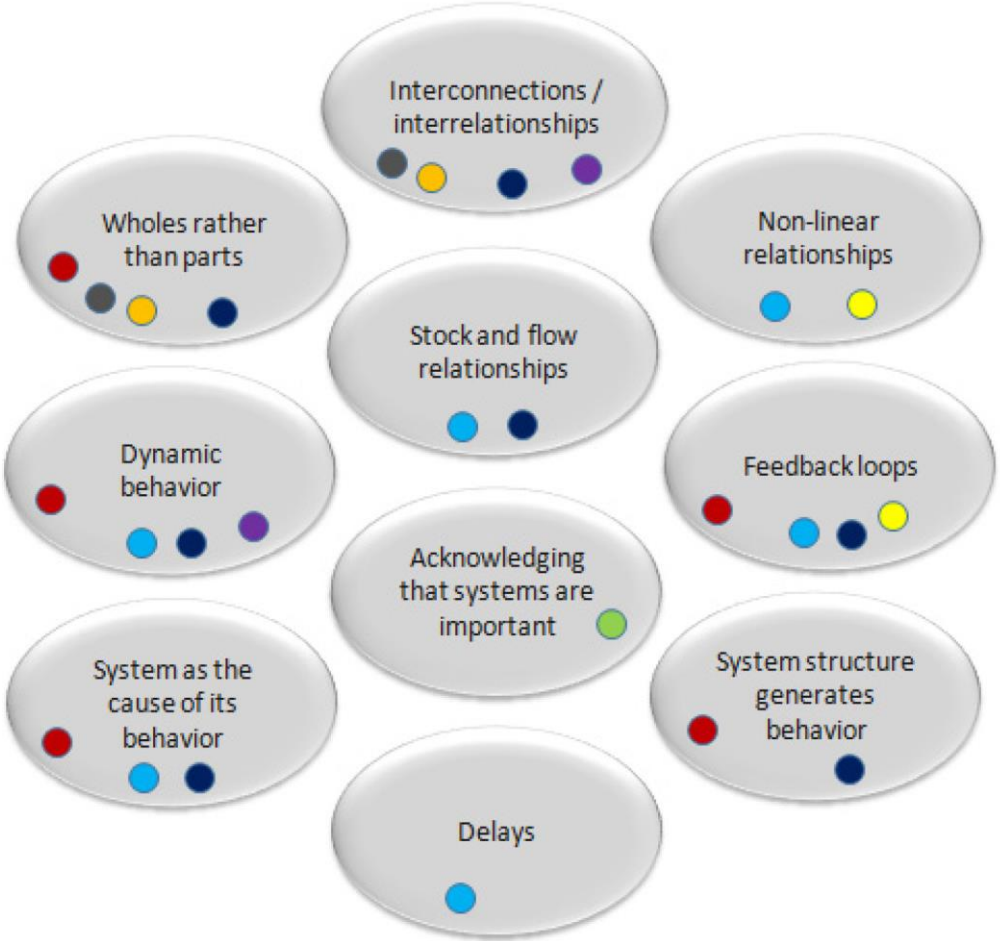
What is systems thinking?

“Systems thinking is a set of synergistic analytic skills used to:

- improve the capability of identifying and understanding systems,
- predicting their behaviors,
- devising modifications to them in order to produce desired effects.

These skills work together as a system.”

Arnold and Wade 2015

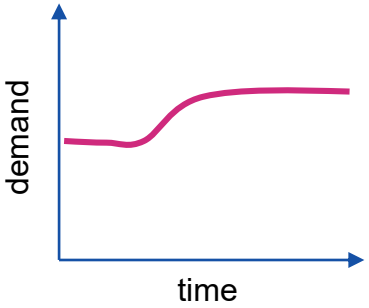


Arnold and Wade 2015

the beer distribution game

- Jay Forrester, MIT



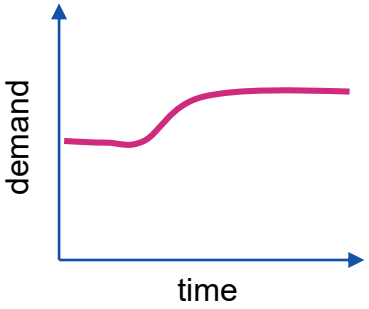


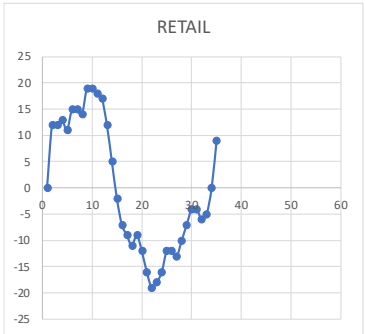


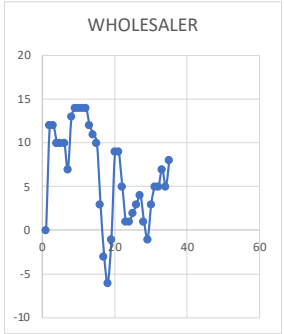
the beer distribution game

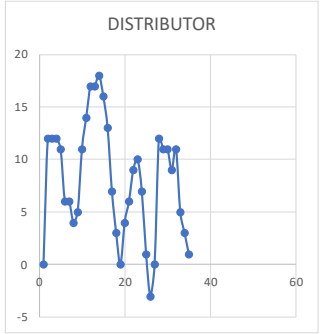
- Jay Forrester, MIT

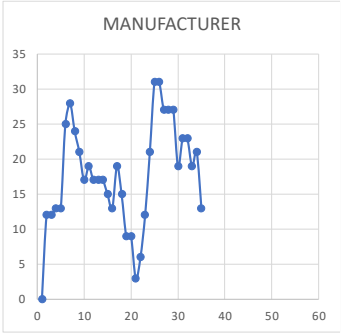












live workshop simulation

The Bullwhip effect in supply chains - the systems perspective



ImageFrom:https://castlevania.fandom.com/wiki/Trevor_Belmont_(animated_series)

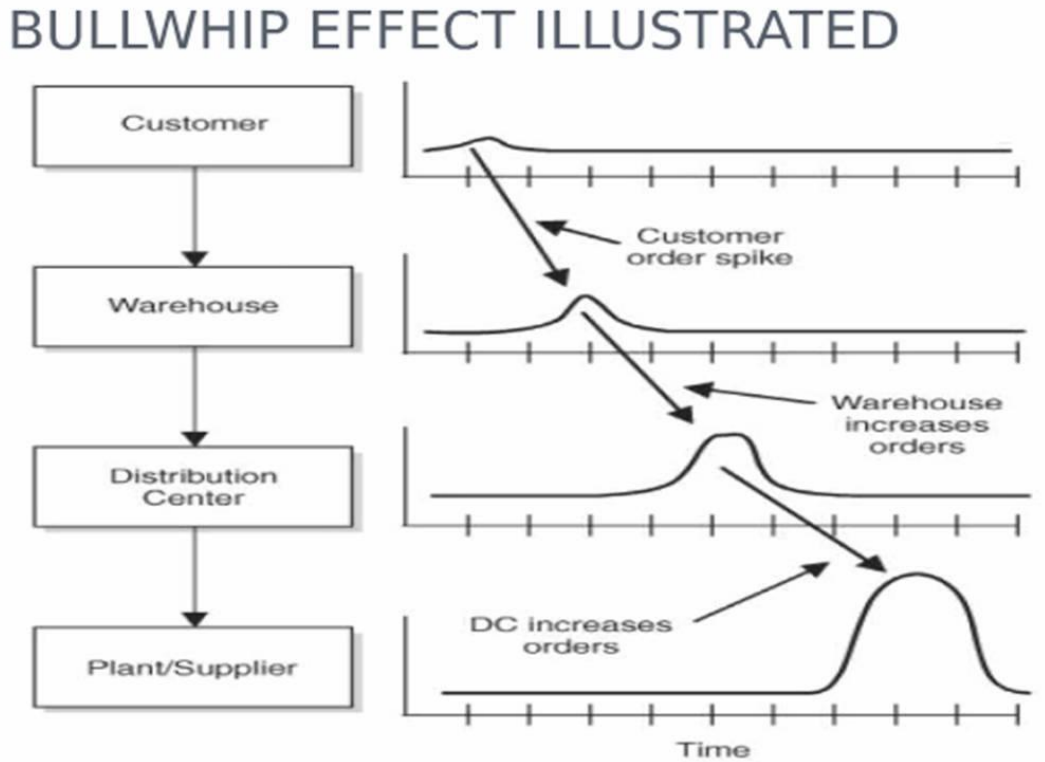


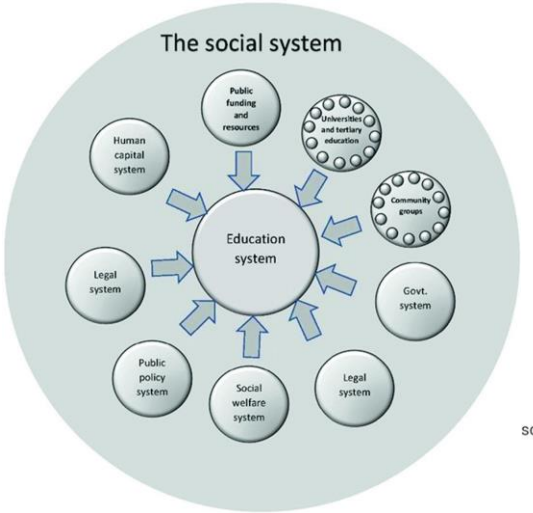
Image from: https://brandalyzer.blog/2016/03/23/the-bullwhip-effect-in-supply-chain/

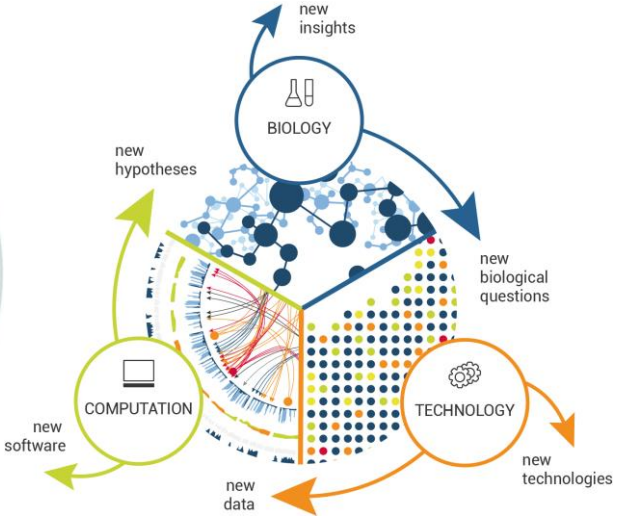
What is system dynamics?

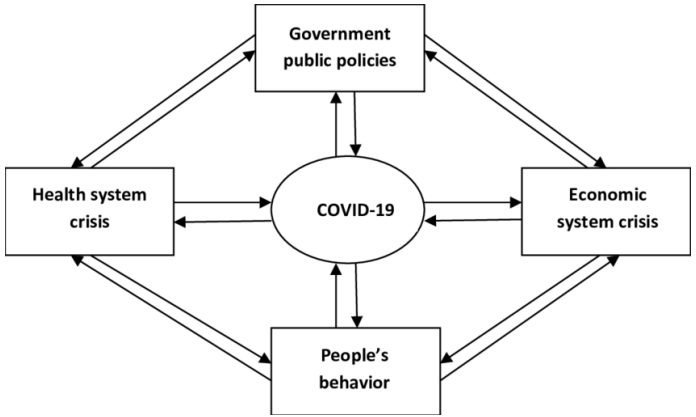
- Computer-aided approach for strategy and policy design.
- Improve understanding of and decision-making related to complex, dynamic systems.
- Simulation modelling based on systems thinking approaches.
- Widely applied in, for example: economics, organisation, ecology, medicine, physiology, healthcare, and more.
- Employing modelling methods such as causal loop diagrams and stocks and flows.

System Dynamics Society 2022; Sterman 2001

When and where is systems thinking and modelling used?



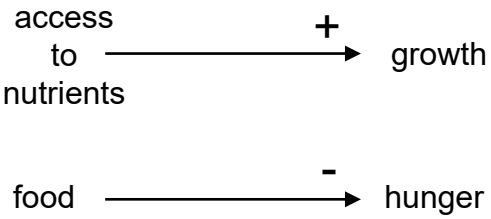




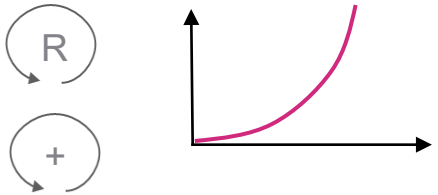


Some basic nomenclature of causal loop diagrams and stock-and-flows models

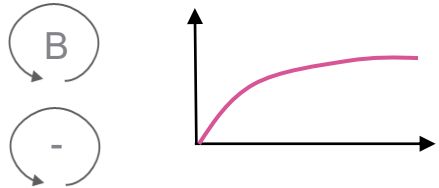
Variables, causal links, polarity



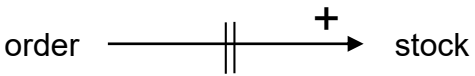
reinforcing feedback loop (R)



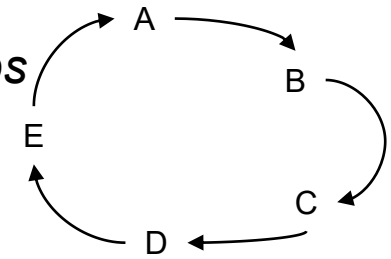
balancing feedback loop (B)



delay



causal loops

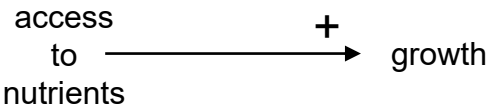


Stocks, flows, sources and sinks

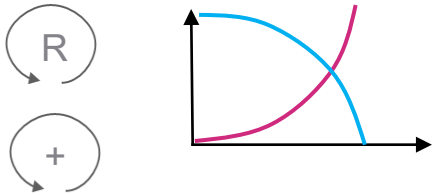


Some basic nomenclature of causal loop diagrams and stock-and-flows models

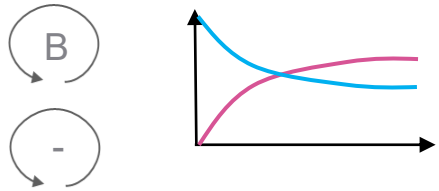
Variables, causal links, polarity



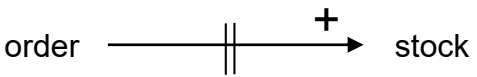
reinforcing feedback loop (R)



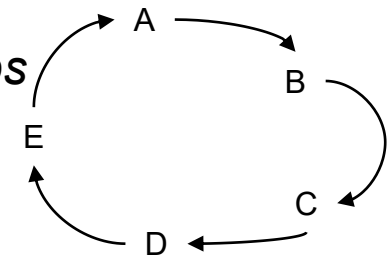
balancing feedback loop (B)

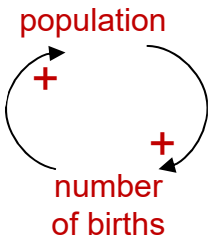


delay

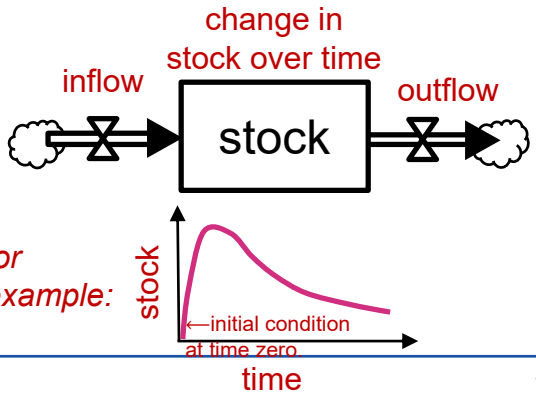


causal loops

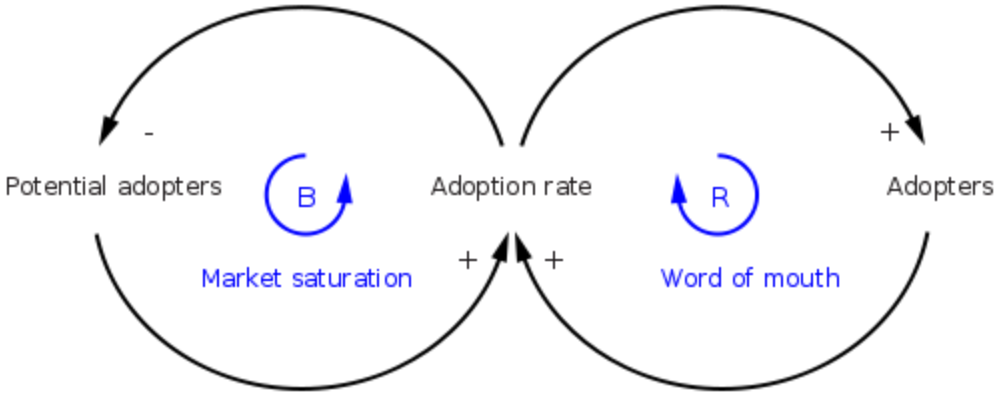




Stocks, flows, sources and sinks

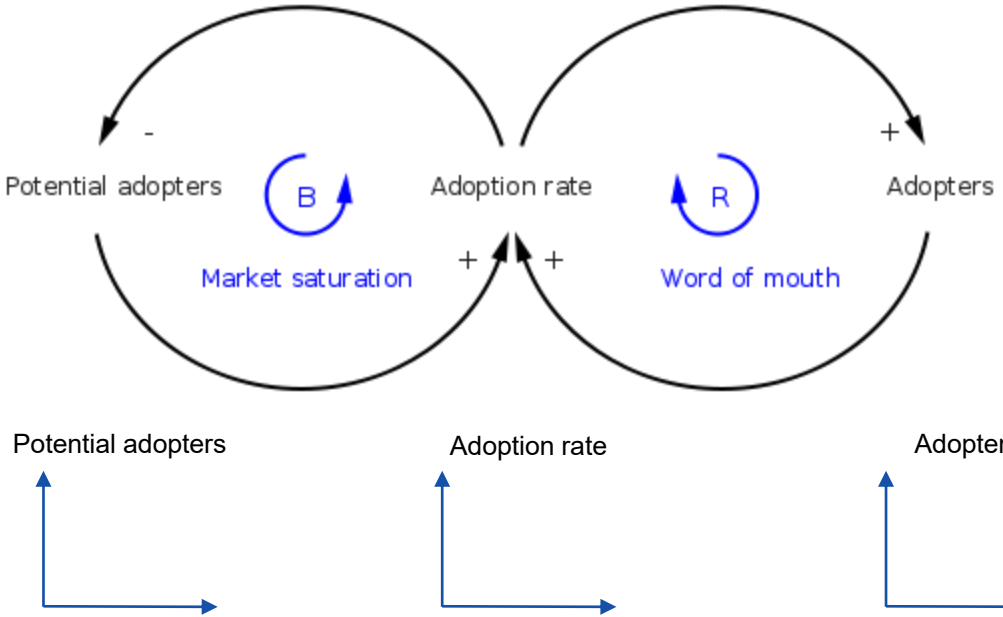


Causal loop diagrams

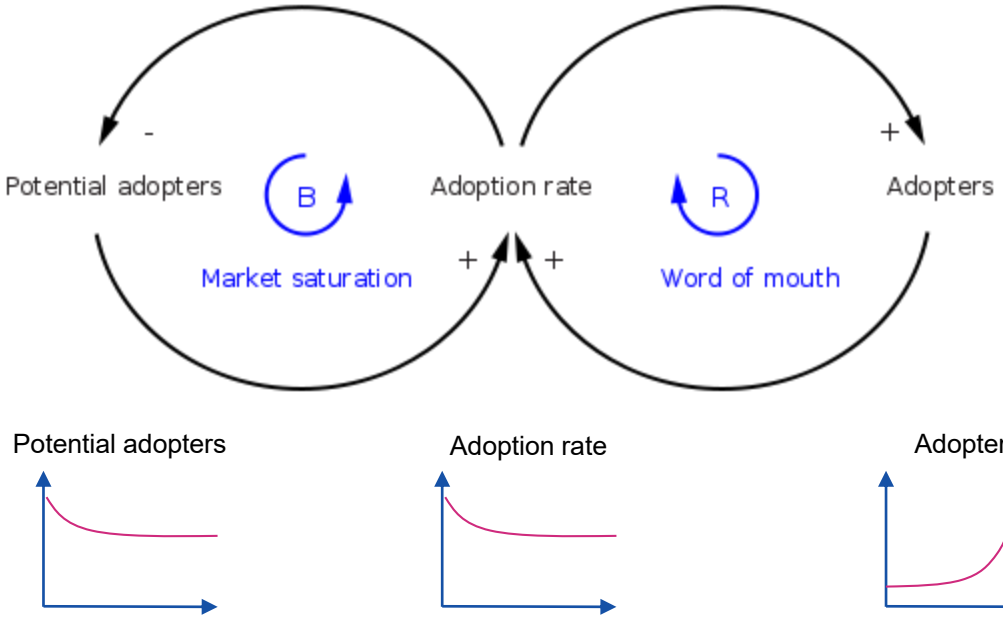


Sterman 2001

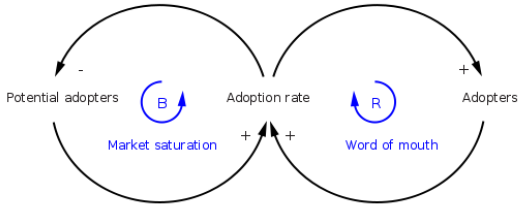
Causal loop diagrams: reference behaviour pattern (RBP)

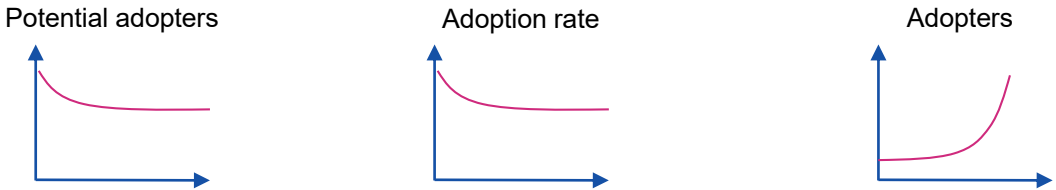


Causal loop diagrams: reference behaviour pattern (RBP)



Causal loop diagrams: reference behaviour pattern (RBP)



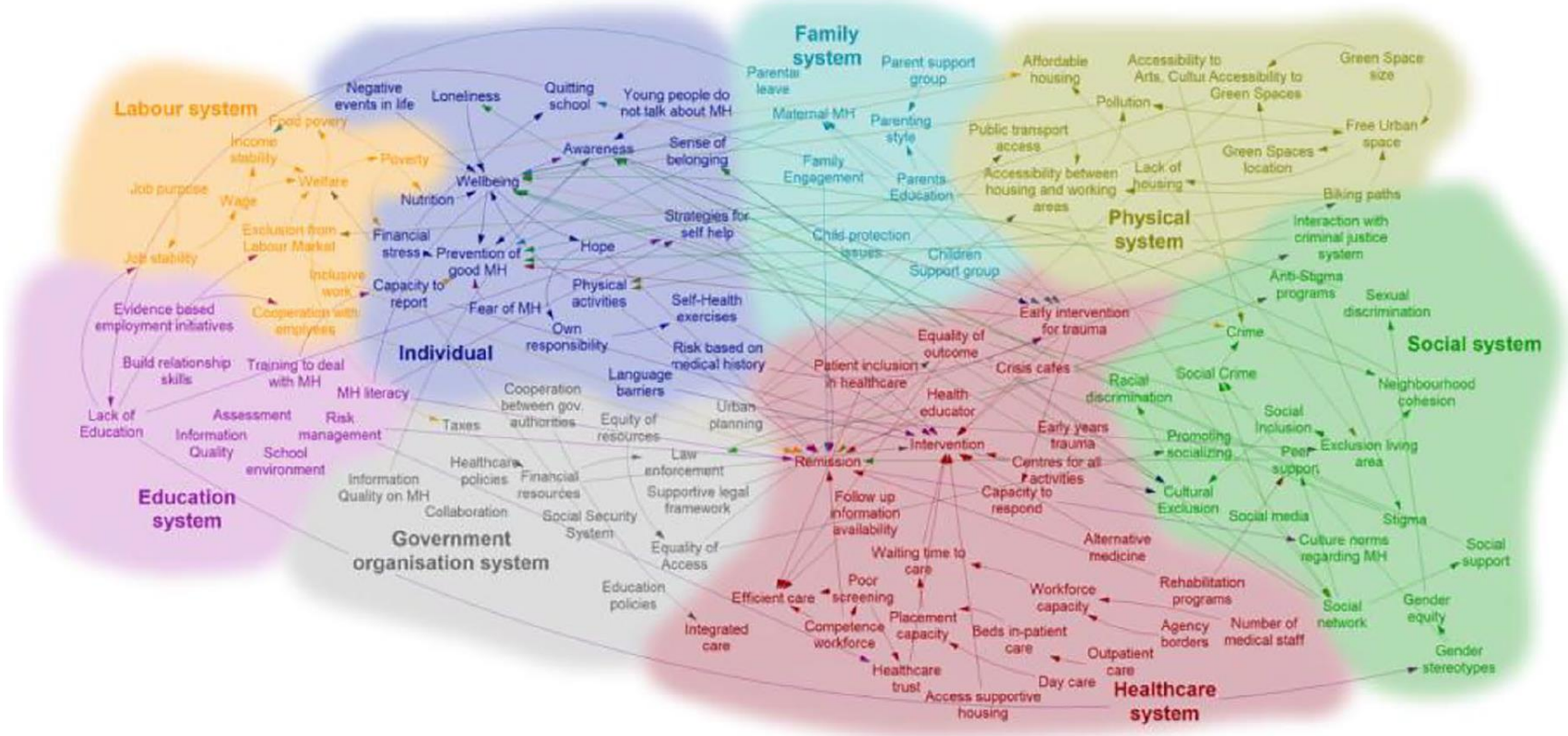


RBP: graphical representation of behaviour over time of variable(s) in the causal loop.

OBP: observed behavioural pattern – historical data.

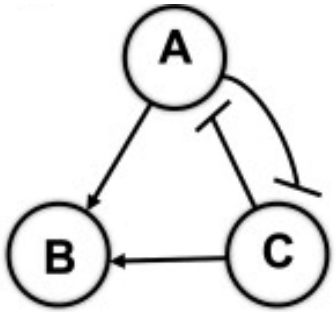
e.g.: what does observed data tell us about adopters over time?

The value of qualitative models: mental health

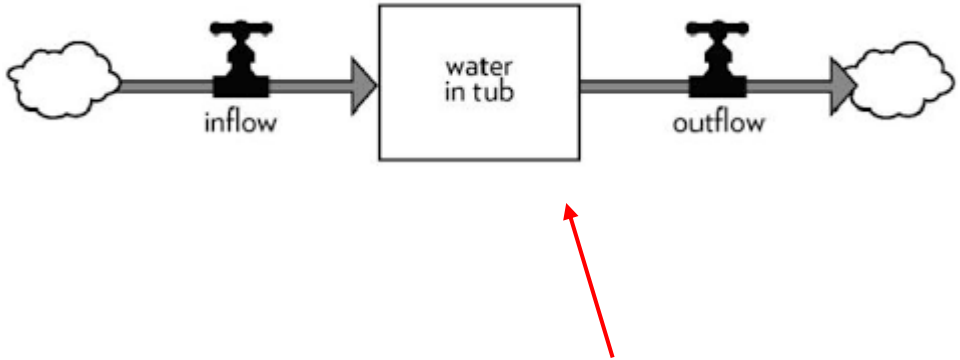


(Moustaid *et al*. 2020)

How do we translate qualitative models into logic, or mathematical expressions for semi-quantitative and quantitative simulations?



Boolean rules:
A* = NOT C
B* = A AND C
C* = NOT A

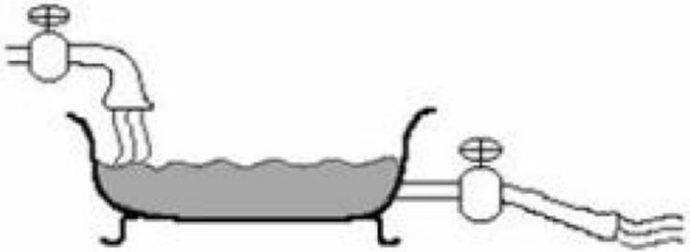


- Boolean simulations
- Graph theory
- Network theory
- Structural equation models

in this course topic we focus on simulations implemented as systems of ordinary differential equations

System dynamics simulation: systems of ordinary differential equations

Hydraulic Metaphor



Stock-Flow Diagram



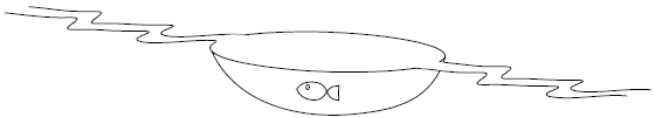
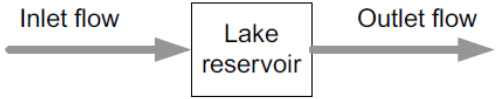
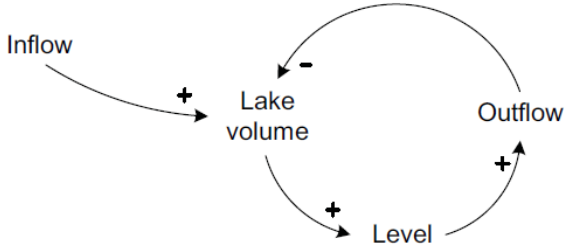
Integral Equation

$$Stock(t) = \int_{t_0}^{t} [Inflow(s) - Outflow(s)]ds + Stock(t_0)$$

Differential Equation

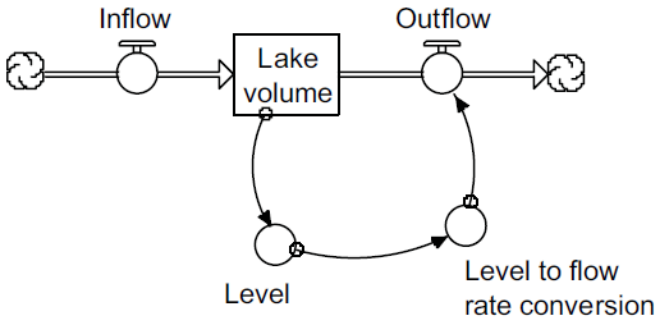
$$d(Stock)/dt = Net\ Change\ Rate = Inflow(t) - Outflow(t)$$

System dynamics simulation: systems of ordinary differential equations

Situation picture	
SFD Stock and flow diagram	
CLD causal loop diagram	

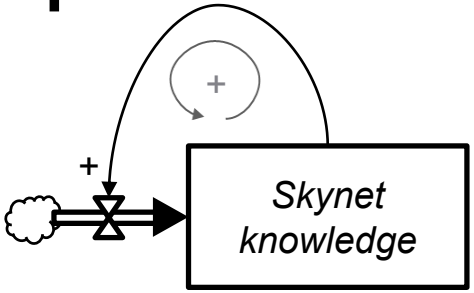
Haraldsson and Svedrup 2020

System dynamics simulation: systems of ordinary differential equations

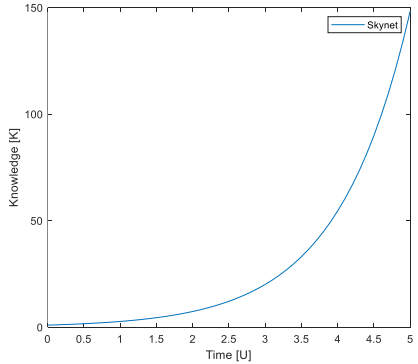
SDTD system dynamics tool diagram	
Equations	Ordinary differential equation: $\frac{dV}{dt} = r_{inflow} - r_{outflow}$ $r_{ouflow} = a \cdot z^n$ $z = \frac{V}{A}$ $A = f(V)$ or, V(t) V = volume r_i = rate of flow z = lake depth a = coefficient A = lake area

The exponential growth model: positive feedback

Example: exponential growth

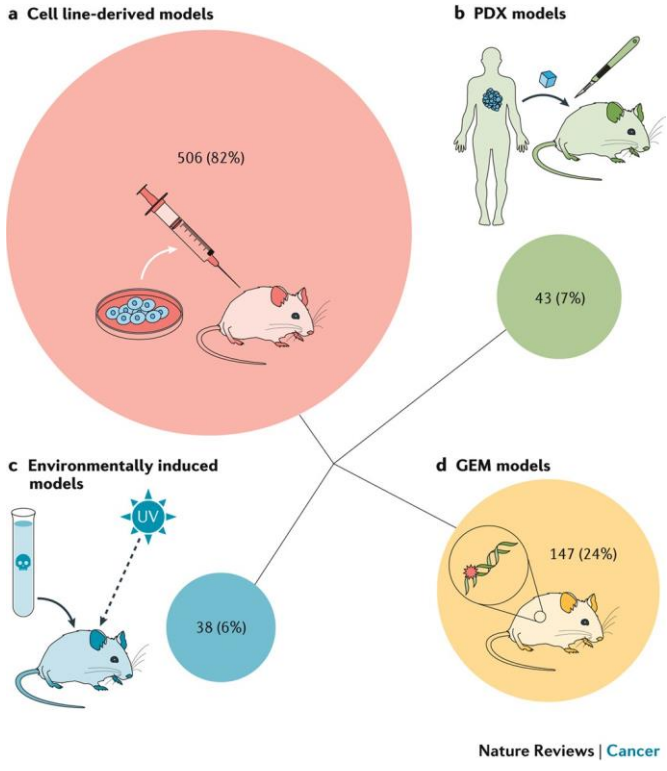


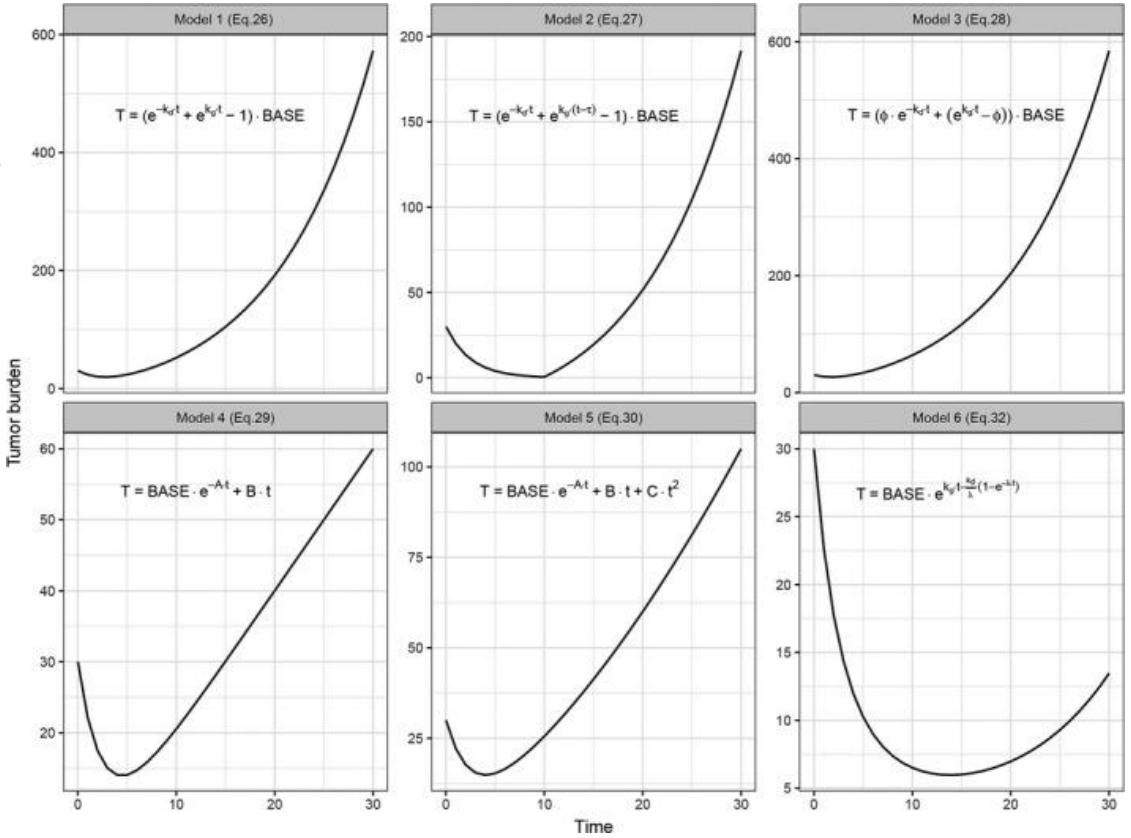
$$\frac{dK_{Skynet}}{dt} = K_{Skynet}\,(t) \cdot k_{learn}$$





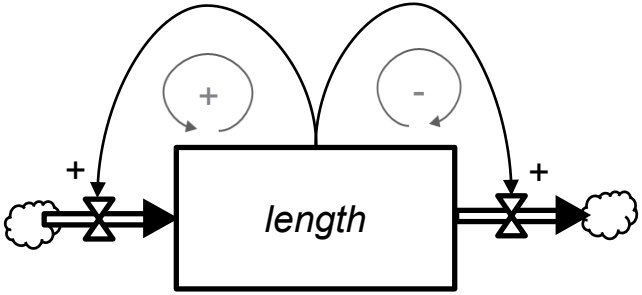
Tumour growth models: modelling in drug development





Gengenbacher *et al*. 2017; Yin *et al*. 2019

Ludwig von Bertalanffy *biological growth*



$$\frac{dL_{animal}}{dt} = L_{animal}(t)^{\alpha} \cdot p - L_{animal}(t) \cdot q$$

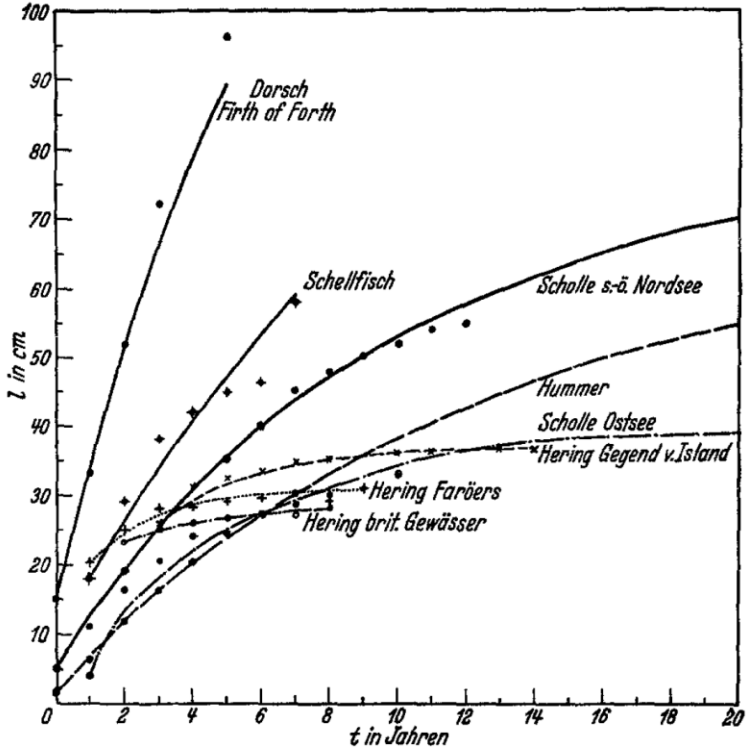


Abb. 4. Längenwachstum von Meerestieren, nach Angaben von PÜTTER (1920), die berechneten Kurven ausgezogen, mit Bezeichnung der empirischen Bestimmungen. (Einige mit anderen zusammenfallende Bestimmungspunkte wurden weggelassen.)

Example: biological growth



systems biology

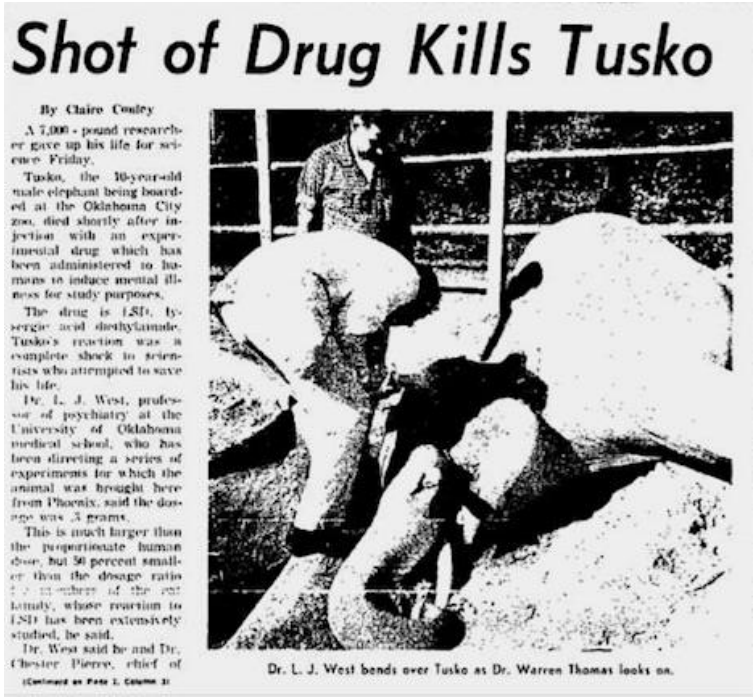
Von Bertalanffy 1934

On the importance of understanding the system: the elephant fallacy

Lysergic Acid Diethylamide: Its Effects on a Male Asiatic Elephant

At 8 A.M. on the second, or experimental, day the LSD was similarly injected. Tusko began trumpeting and rushing around the pen, a reaction not unlike the one he had shown the day before. However, this time his restlessness appeared to increase for 3 minutes after the injection; then he stopped running and showed signs of marked incoordination. His mate (Judy, a 15-year-old female) approached him and appeared to attempt to support him. He began to sway, his hindquarters buckled, and it became increasingly difficult for him to maintain himself upright. Five minutes after the injection he trumpeted, collapsed, fell heavily onto his right side, defecated, and went into *status epilepticus*. The limbs

Downloaded fro



Shot of Drug Kills Tusko

By Claire Conley

A 7,000-pound researcher gave up his life for science Friday.

Tusko, the 10-year-old male elephant being boarded at the Oklahoma City zoo, died shortly after injection with an experimental drug which has been administered to humans to induce mental illness for study purposes.

The drug is LSD, lysergic acid diethylamide. Tusko's reaction was a complete shock to scientists who attempted to save his life.

Dr. L. J. West, professor of psychiatry at the University of Oklahoma medical school, who has been directing a series of experiments for which the animal was brought here from Phoenix, said the dosage was .3 grams.

This is much larger than the proportionate human dose, but 50 percent smaller than the dosage ratio [illegible] of the rat family, whose reaction to LSD has been extensively studied, he said.

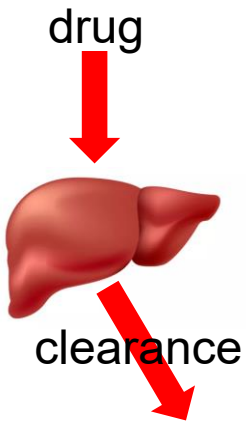
Dr. West said he and Dr. Chester Pierce, chief of

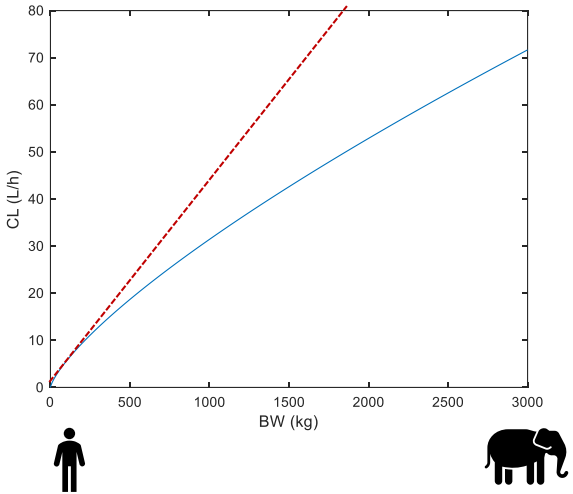
(Continued on Page 2, Column 3)

Dr. L. J. West bends over Tusko as Dr. Warren Thomas looks on.

(West *et al*. 1962)

On the importance of understanding the system: the elephant fallacy





BW: bodyweight
CL: clearance, elimination flow of drug from the body (volume of blood cleared from drug over time, e.g. L/h).
AUC: area-under-the concentration-time curve.

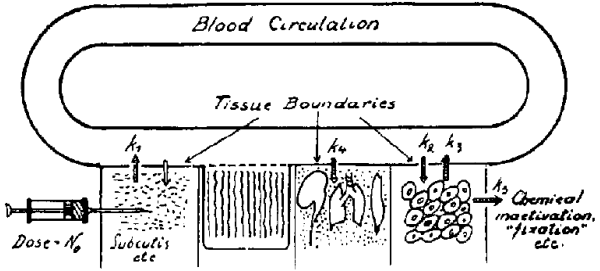
$$\alpha_j = \alpha_i \cdot \left(\frac{BW_j}{BW_i}\right)^{\beta}$$

$e.g.$:

$$CL_j = CL_i \cdot \left(\frac{BW_j}{BW_i}\right)^{0.75}$$

$$Dose = AUC/CL$$

Torsten Teorell *pharmacokinetics*



Local	Drug depot	Blood + equivalent Blood vol	Kidney etc elimination	Tissues	Tissue inactivation
Symbol	D	B	K	T	I
Amount	x	y	u	z	w
Volume	V_1	V_2	—	V_3	—
Concentration	x/V_1	y/V_2	—	z/V_3	—
Perm. coeff	k_1'	—	k_4'	k_2'	—
Velocity constant out	$k_1 = k_1'/V_1$	—	$k_4 = k_4'/V_2$	$k_3 = k_2'/V_2$	k_5
Velocity constant in	neglected	—	not existing	$k_2 = k_2'/V_2$	—
Name of process	Resorption	—	Elimination	Tissue take up — — output	Inactivation

FIG. 1

Scheme of the Concept of Drug Distribution used in this paper. Instead the *injection* pictured in the figure, the administration of the drug depot can be made *per os*, *per rectum*, *by inhalation*, etc.

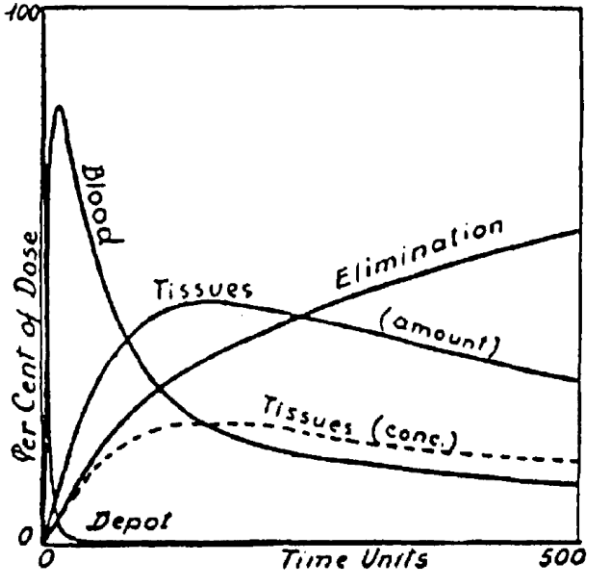


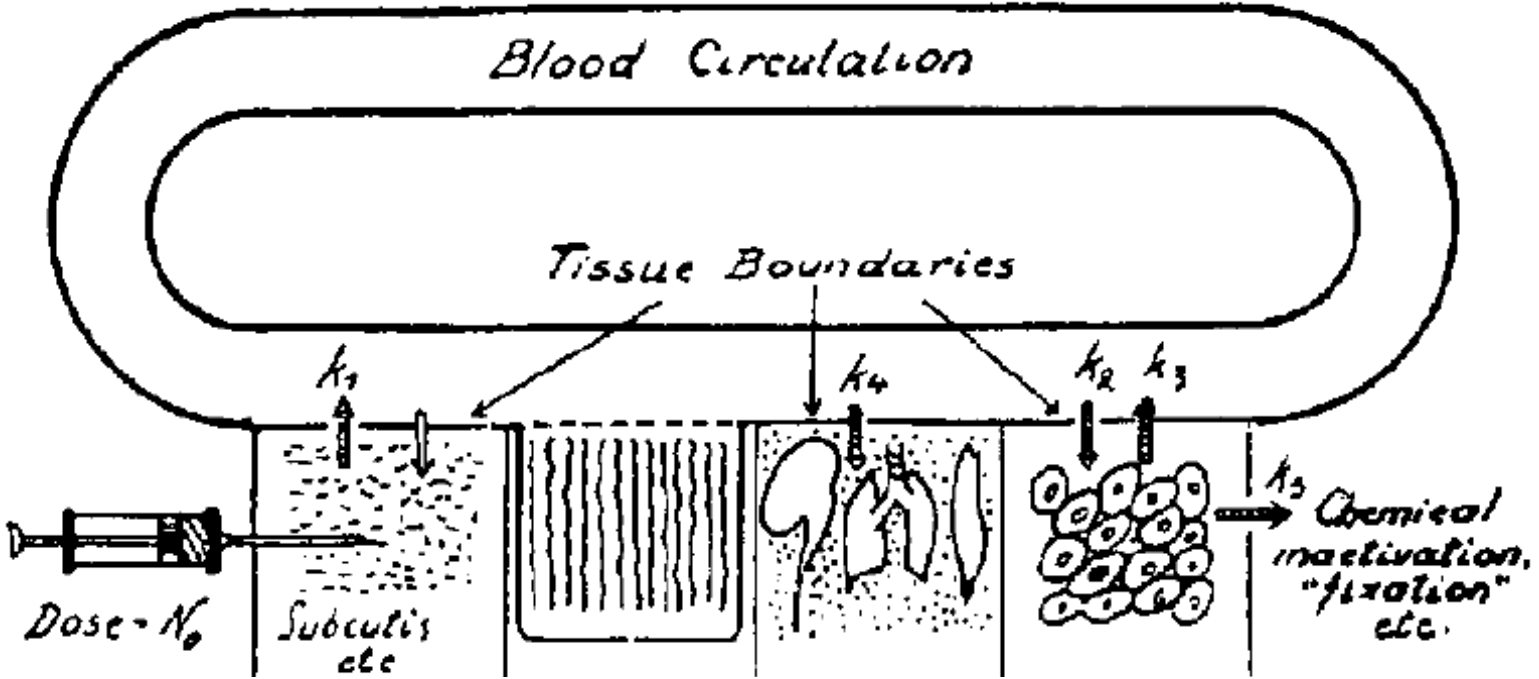
FIG. 3

Typical Case of Extravascular Administration in the absence of tissue inactivation.

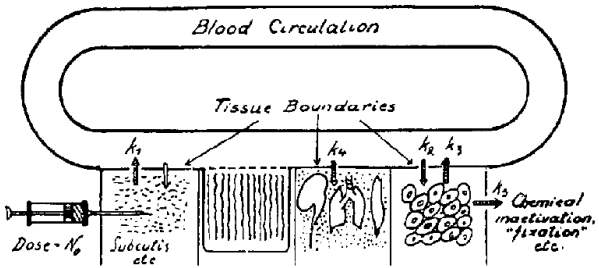


systems pharmacology, physiologically-based pharmacokinetics

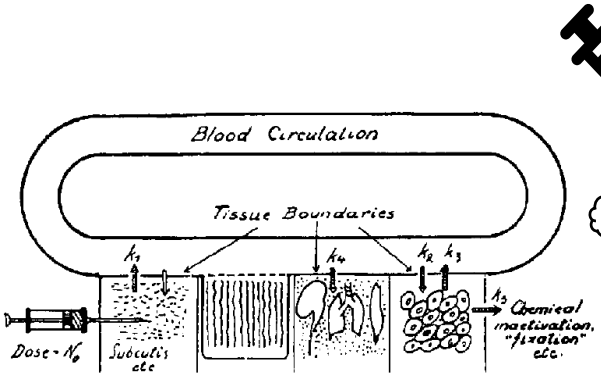
Pharmacokinetic model

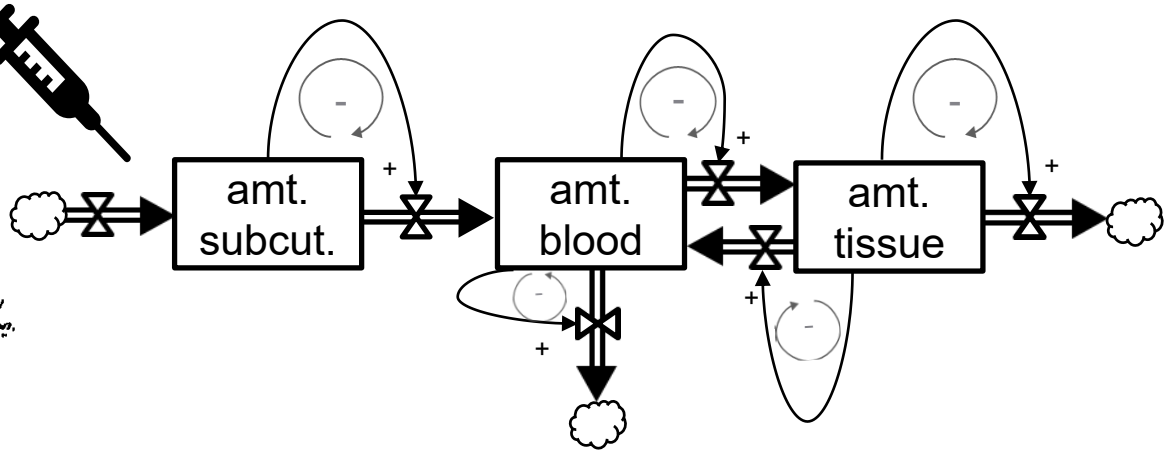


Pharmacokinetic model



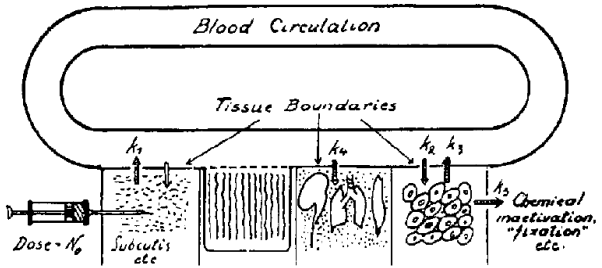
Pharmacokinetic model

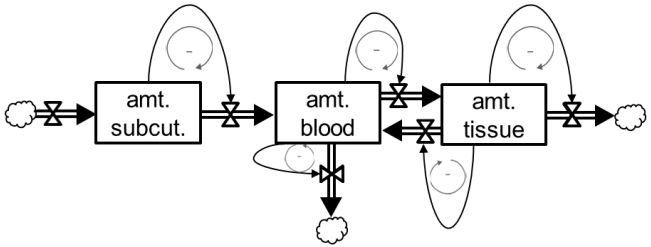




Pharmacokinetic model

Example: Teorell PK model





$$\frac{dA_{sc}}{dt} = -A_{sc}(t) \cdot k_1, \qquad A_{sc}(t=0) = Dose,$$

$$\frac{dA_{bl}}{dt} = A_{sc}(t) \cdot k_1 - A_{bl}(t) \cdot (k_2 + k_4) + A_{tis}(t) \cdot k_3$$

$$\frac{dA_{tis}}{dt} = A_{bl}(t) \cdot k_2 - A_{tis}(t) \cdot (k_3 + k_5)$$

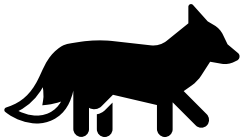
$$\frac{dA_{elim}}{dt} = A_{bl}(t) \cdot k_4$$

$$\frac{dA_{inac}}{dt} = A_{tis}(t) \cdot k_5$$

Systems pharmacology

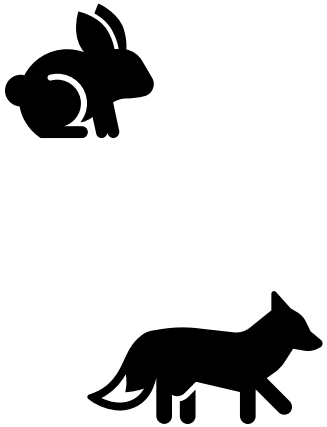
Predator-prey population

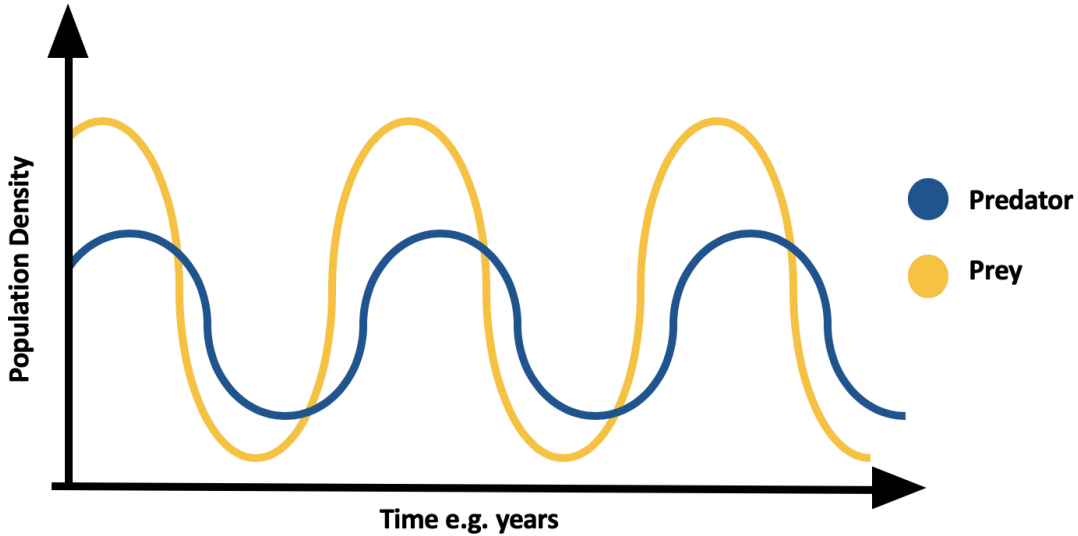






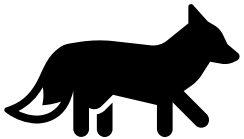
Lotka-Volterra: predator-prey model

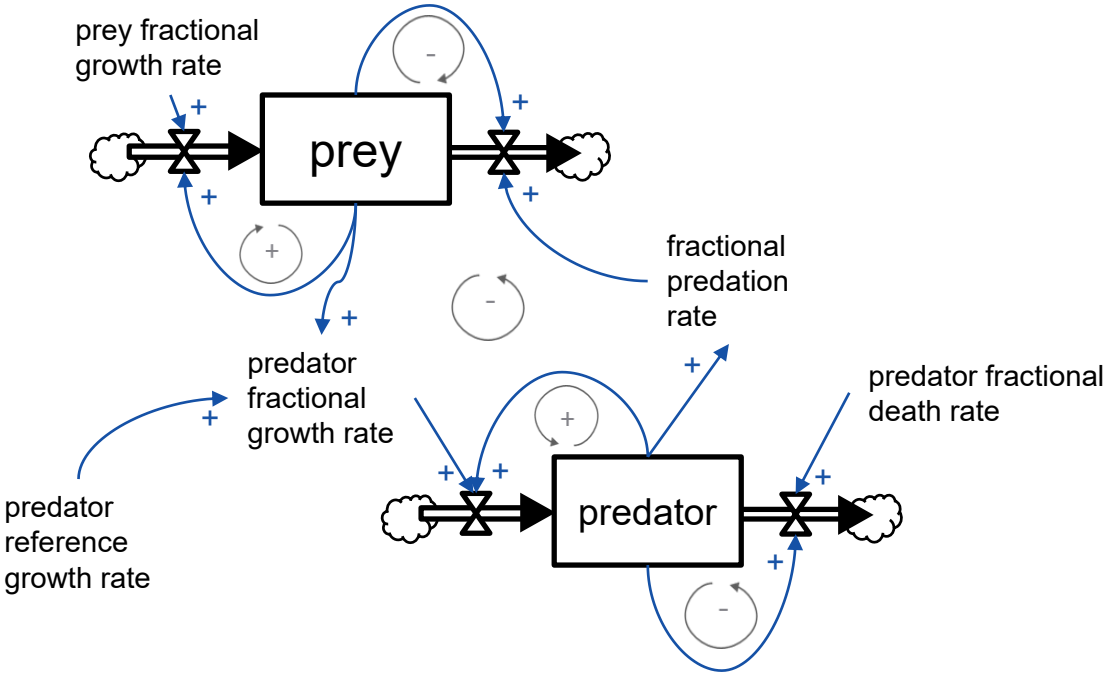




Lotka-Volterra: predator-prey model

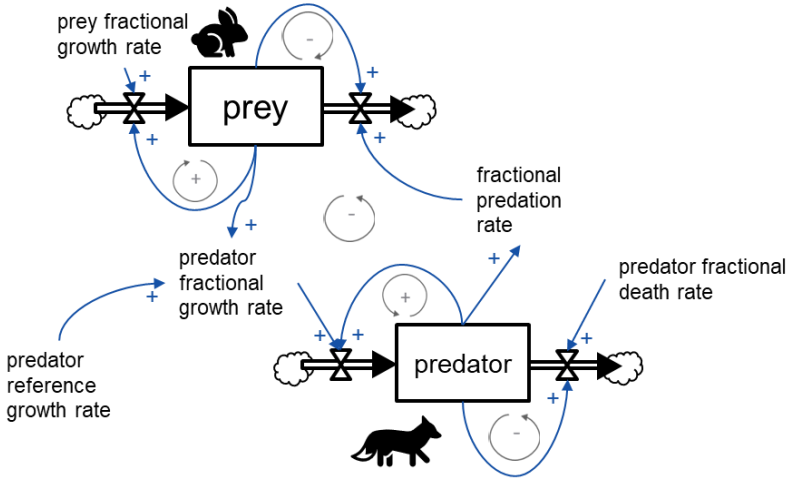






Lotka-Volterra: predator-prey model

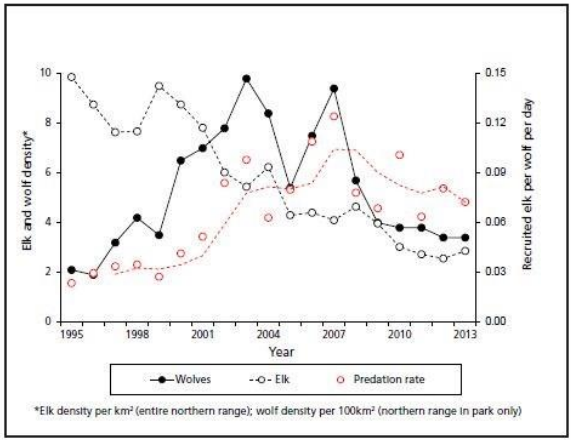
Example:
Lotka Volterra
model

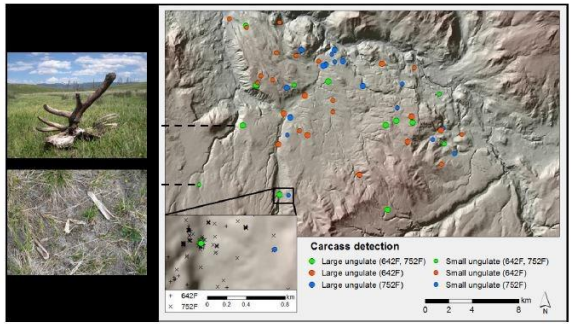


$$\frac{dA_{prey}}{dt} = A_{prey}(t) \cdot \propto -A_{prey}(t) \cdot A_{pred}(t) \cdot \beta, \qquad A_{prey}(t=0) = n,$$

$$\frac{dA_{pred}}{dt} = A_{pred}(t) \cdot A_{prey}(t) \cdot \sigma - A_{pred}(t) \cdot \gamma, \qquad A_{pred}(t=0) = m,$$

Yellowstone Park ecosystem: the systems perspective





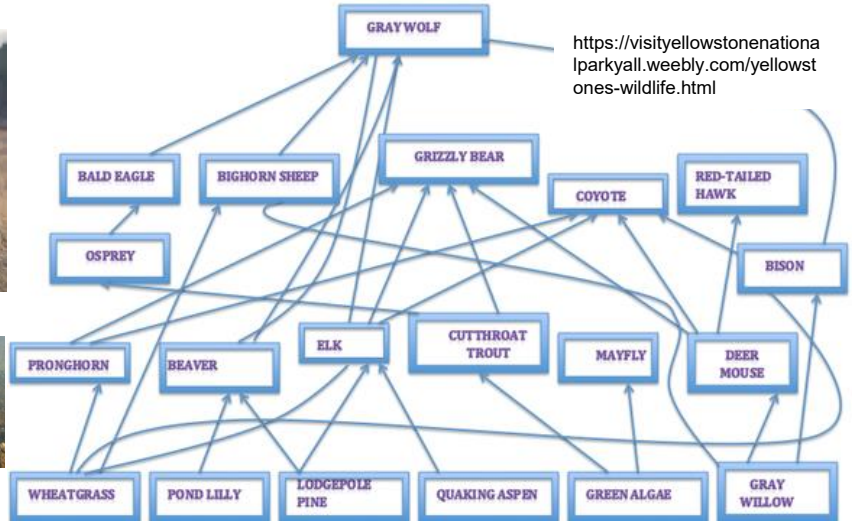








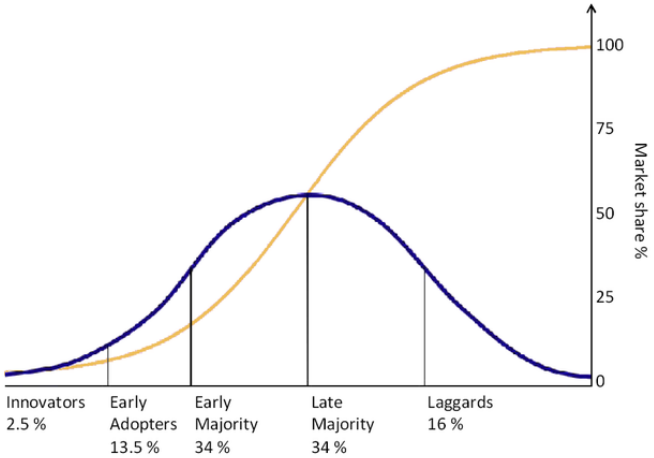


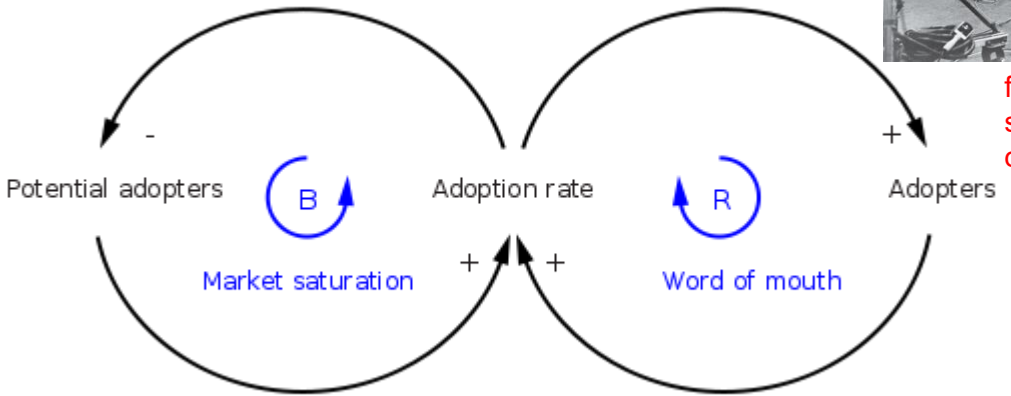


https://visityellowstonenationalparkyall.weebly.com/yellowstones-wildlife.html

Metz *et al*. 2016; Yellowstone, National Park Service

Jay Forrester *system dynamics of adopters*





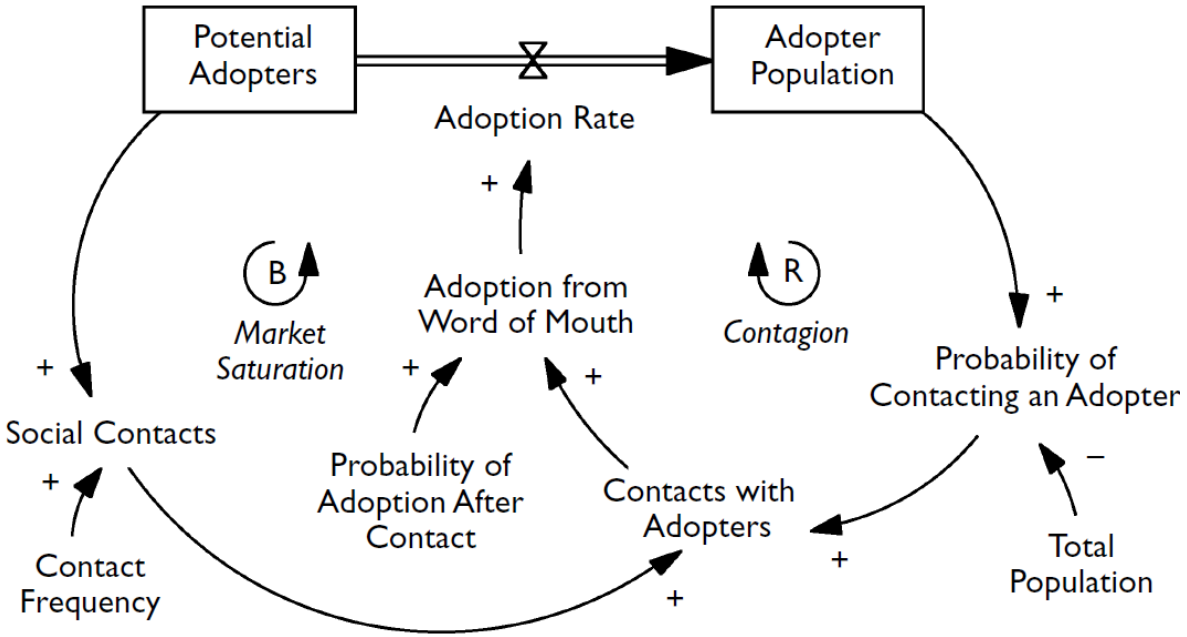


founder of system dynamics

potential adopters → adopters

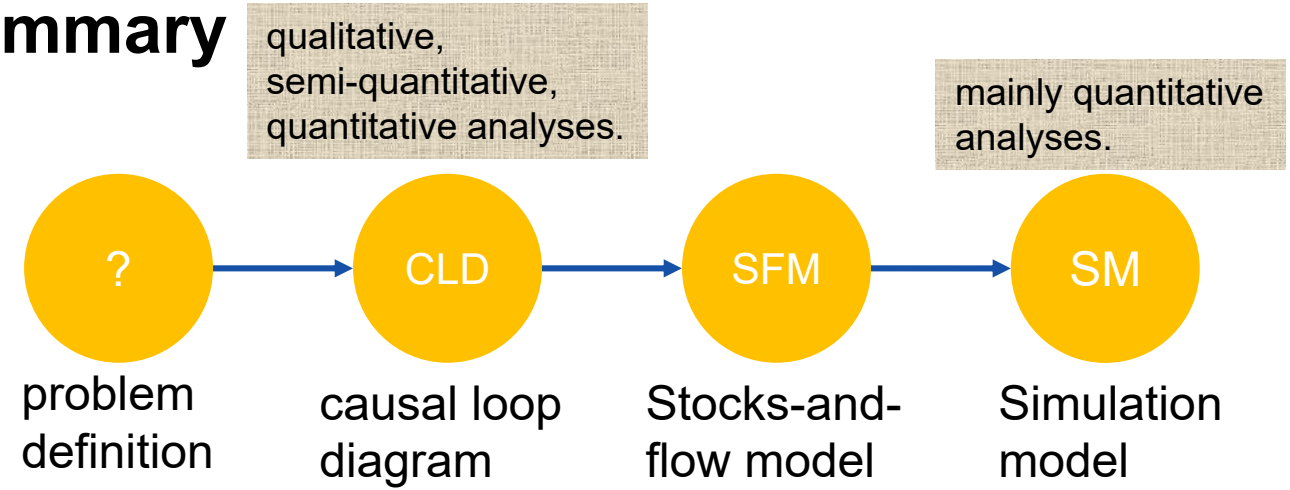
Sterman 2001

Example:
Forrester
Adopter
model



Sterman 2001

Summary

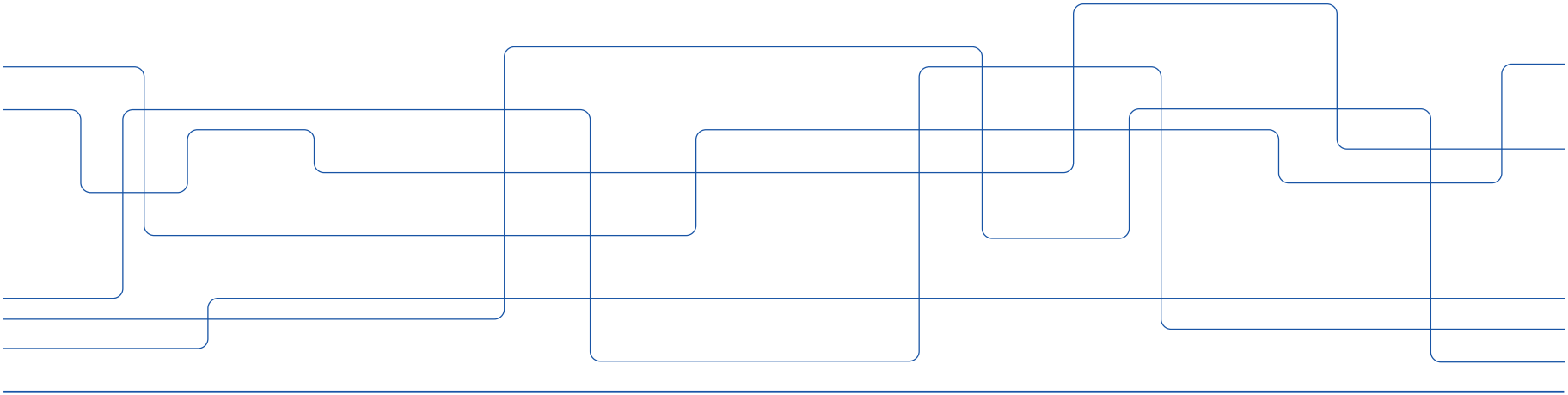


- Familiarity and understanding of the basic principles and nomenclature of system dynamic modelling and simulation.
- Powerful approach for systematic study and change work in systems.
- How can we go from a problem to a model that is fit-for-purpose?
- In the next session we will focus on good practices in system dynamics



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OF TECHNOLOGY

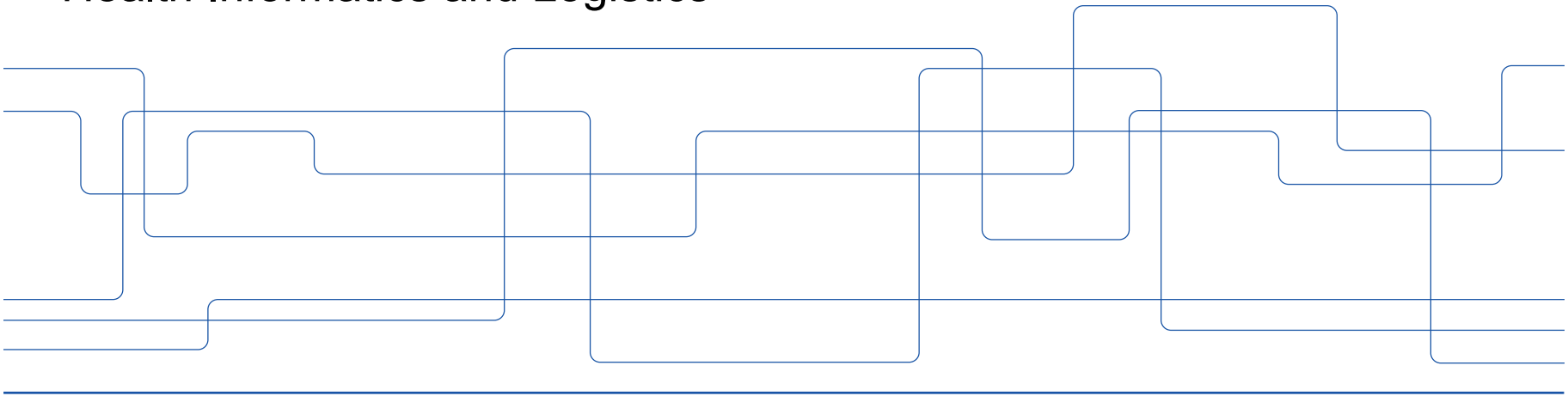
contact: darwich@kth.se



CM2014

System dynamic modelling and simulation
Lecture 2: good practices

Adam S. Darwich
Associate Professor of Health Systems Engineering
Health Informatics and Logistics



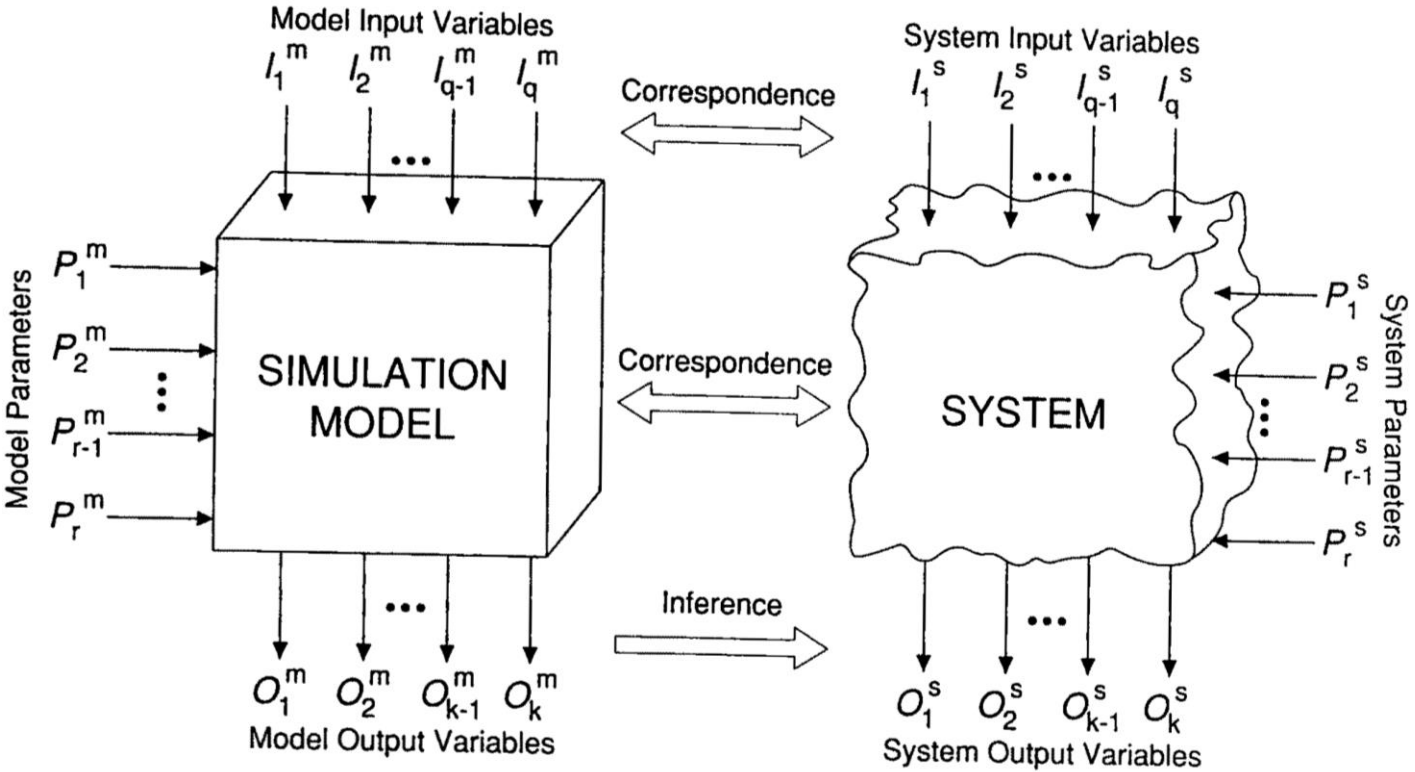


Figure 1. Model and System Characteristics.

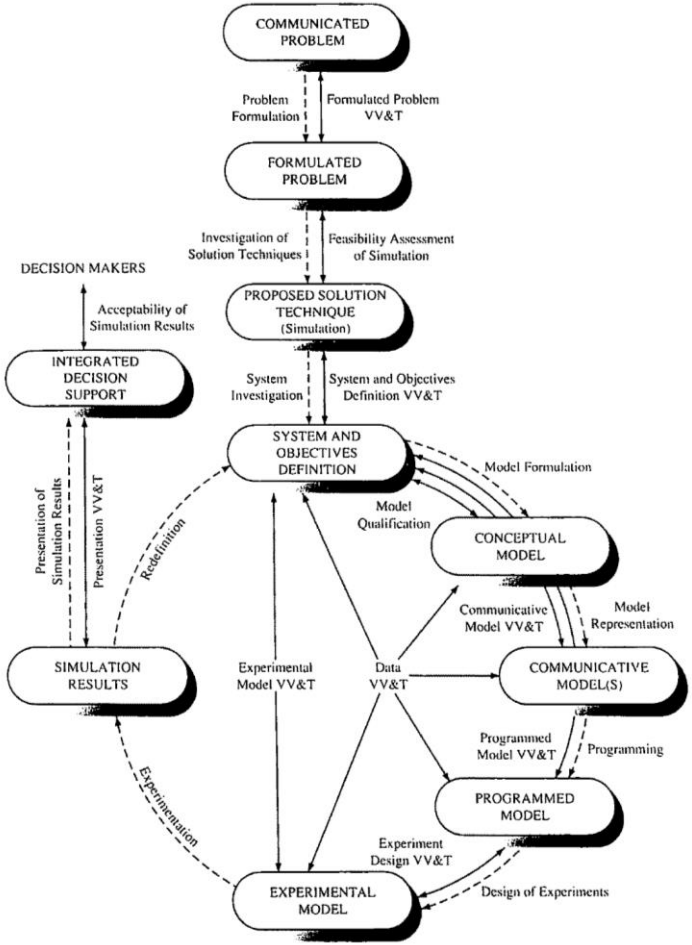
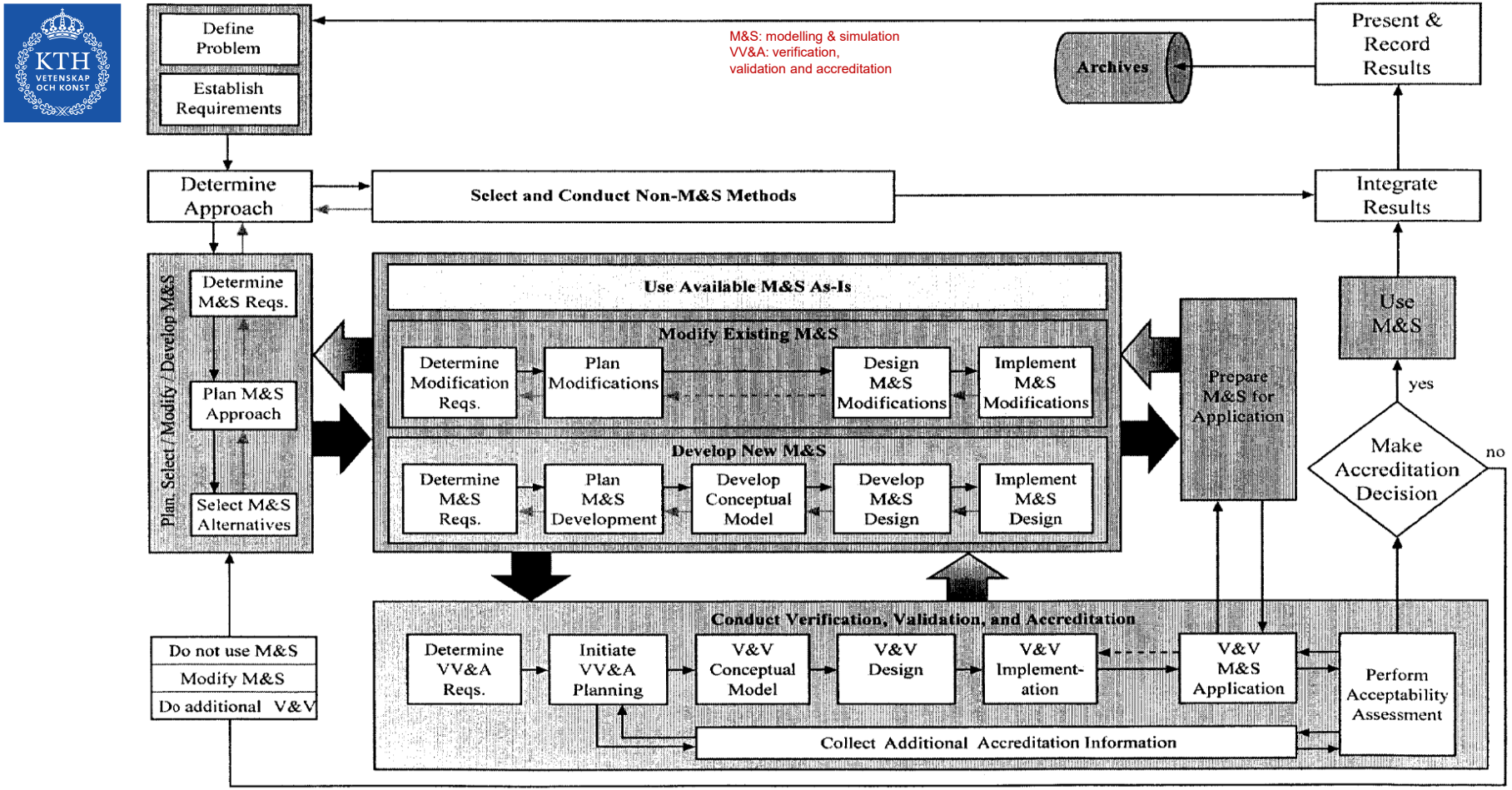


Figure 2. The life cycle of a simulation study.

V&VT: verification, validation and testing


M&S: modelling & simulation
VV&A: verification,
validation and accreditation
Define Problem
Establish Requirements
Archives
Present & Record Results
Determine Approach
Select and Conduct Non-M&S Methods
Integrate Results
Plan, Select / Modify / Develop M&S
Determine M&S Reqs.
Plan M&S Approach
Select M&S Alternatives
Use Available M&S As-Is
Modify Existing M&S
Determine Modification Reqs.
Plan Modifications
Design M&S Modifications
Implement M&S Modifications
Develop New M&S
Determine M&S Reqs.
Plan M&S Development
Develop Conceptual Model
Develop M&S Design
Implement M&S Design
Prepare M&S for Application
Use M&S
yes
Make Accreditation Decision
no
Conduct Verification, Validation, and Accreditation
Determine VV&A Reqs.
Initiate VV&A Planning
V&V Conceptual Model
V&V Design
V&V Implement-ation
V&V M&S Application
Perform Acceptability Assessment
Collect Additional Accreditation Information
Do not use M&S
Modify M&S
Do additional V&V

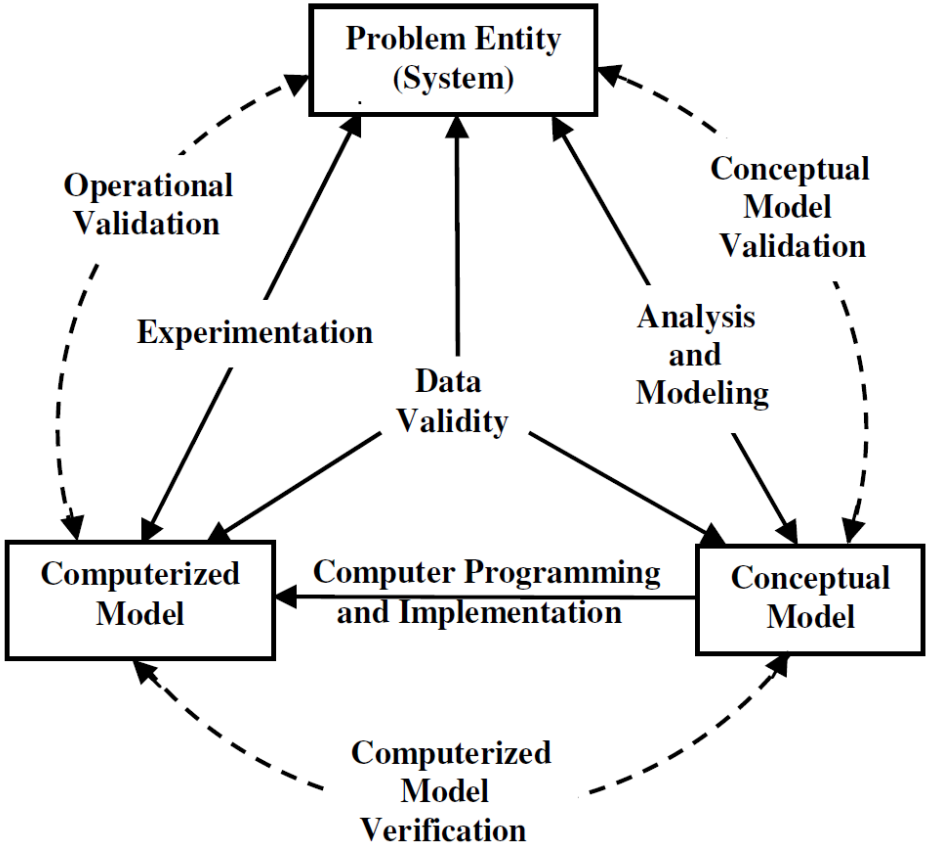


Figure 2: Simplified Version of the Modeling Process

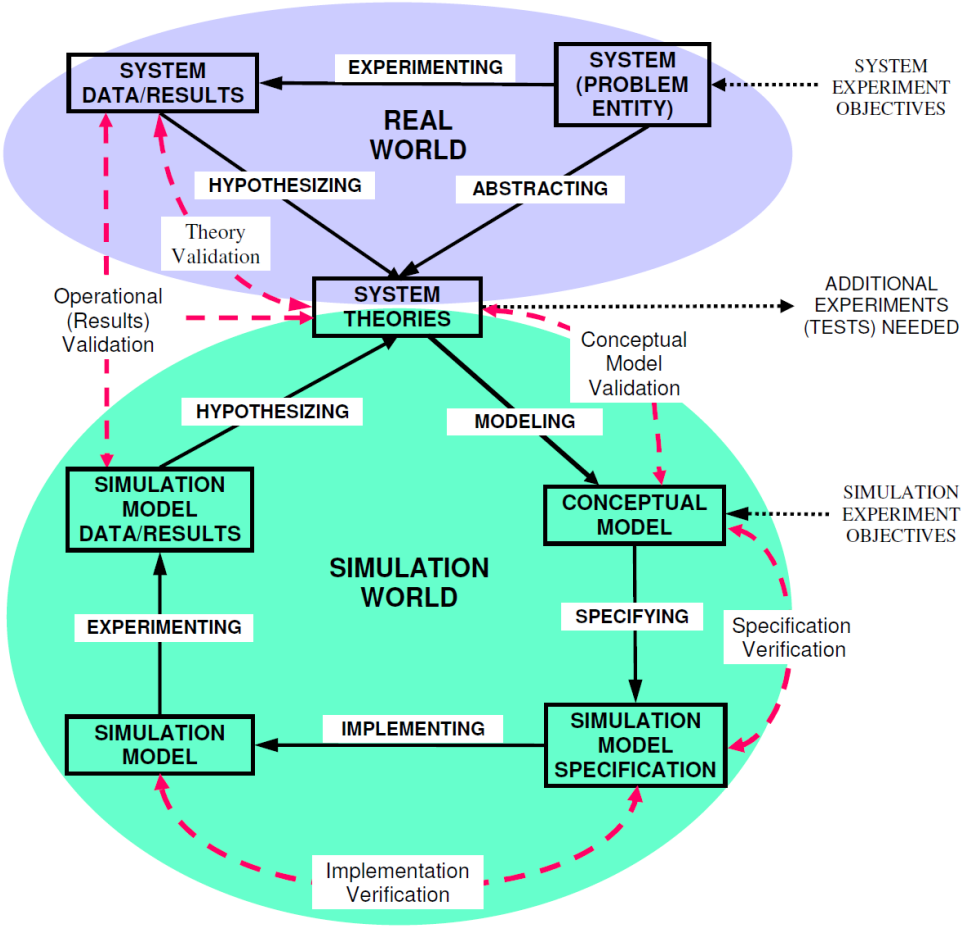


Figure 3: Real World and Simulation World Relationships with Verification and Validation

Table 1. Summary of the number of statements of best practice elicited, broken down by modeling stage, perceived importance, and extent of agreement among the participants

		Stages of System Dynamics Modeling Process							
Importance		Problem Identification and Definition	System Conceptualization	Model Formulation	Model Testing and Evaluation	Model Use, Implementation, and Dissemination	Design of Learning Strategy/Infrastructure	Number of statements	Percentage
Highest	*High Agreement*	5	3	3	1	1	1	14	7
	Low Agreement	2	3	2	3	1	2	13	7
High		5	3	3	3	1	2	17	9
Average		5	3	6	4	6	4	28	14
Low		32	26	28	21	11	8	126	64
Total		49	38	42	32	20	17	198	100

Martinez-Moyano and Richardson 2013

Formulate problem and boundaries

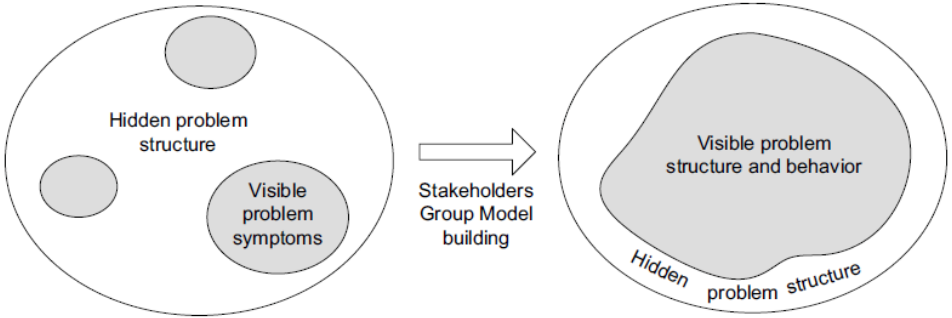


Figure 3.1. How the stakeholders clarify symptoms and the underlying structure of the problem.

Haraldsson and Svedrup 2020

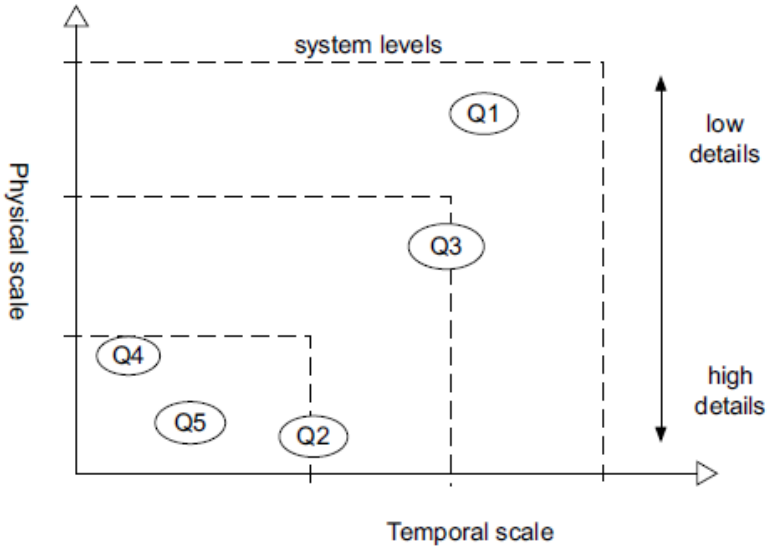
- Engage with experts.
- Define the problem, e.g.:
  - What are the symptoms?
  - What are the causalities?
- Map the problem with mind-maps, causal loop diagrams, stocks-and-flows.
- Define system boundaries according to the group's understanding of the problems, these can change!

Define research questions

Q1. What role does the local aquifer have for the regional water security?

Q2. What are the benefits of the water quality from the aquifer for the local municipality?

Q3. What are the impacts of urban encroachment on the aquifer?

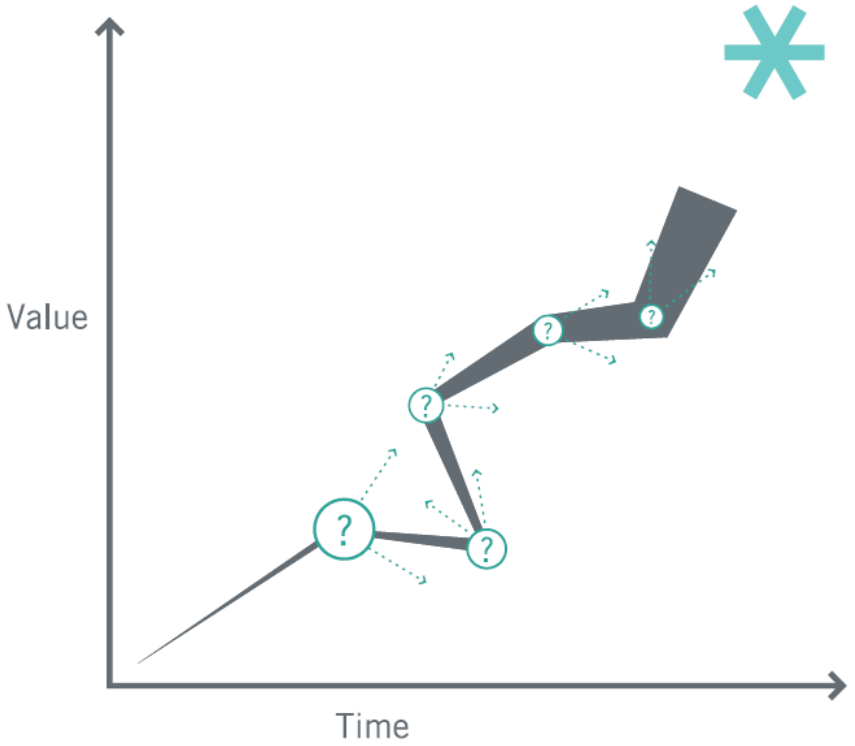


- Explicitly define the aims and objectives with the modelling activity.
- This can redefine systems boundaries.
- Focus and prioritise – what are the most important questions for the stakeholder group.
- The purpose is also important for determining model validity and accreditation/fit-for-purpose/operational validity/credibility.
- Mapping variables, stakeholders, data. Data impacts feasibility and new data can challenge validity.

Haraldsson and Svedrup 2020; Balci 1998; Sargent 2010

Defining a system: a design process

An iterative and adaptive strategy moves towards a North Star framed around possible outcomes. An iterative approach enables a series of pivot points, which allows the mission to navigate through complex terrain, as well as taking advantage of new innovations as they occur, or responding to challenges as they happen. It enables different scenarios to be evaluated as the mission develops, at these pivot points, rather than trying to predict eventualities in advance. As the project progresses, and the mission theme's North Star is in sight, the decisions get easier, smaller. Conversely, while the start of the process is low-risk; as it develops, the value produced by the portfolios of projects increases.—Mike Tyson



Hill *et al*. 2022

Defining a system: a design process

1 Linear, waterfall, predictive, fund-and-forget



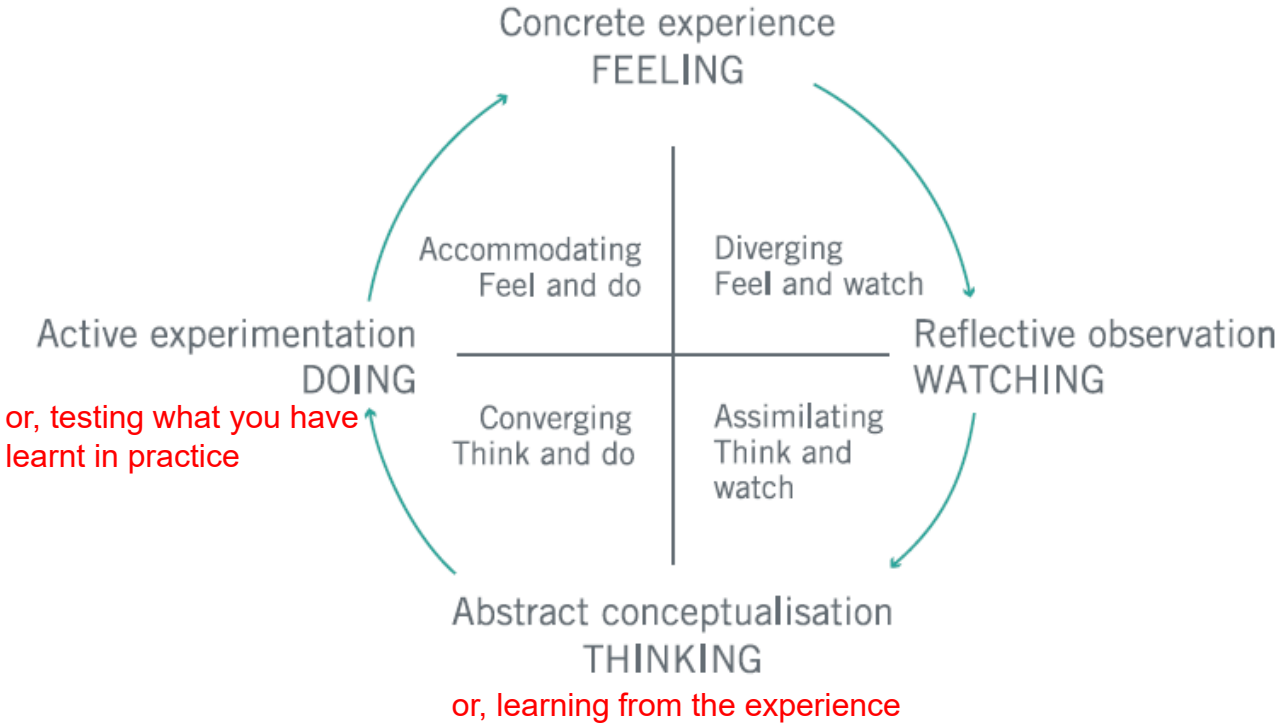
2 Parallel, iterative, adaptive, engaged learning



These two models provide very different modes of development. The first describes a typical linear model, in which delivery follows policy and planning. Learning, and adaptation, struggles to swim back upstream. The second model allows a more engaged interplay between delivery modes, with policy learning from regular feedback loops.

Hill *et al*. 2022

Defining a system: a design process



or, testing what you have learnt in practice

or, learning from the experience

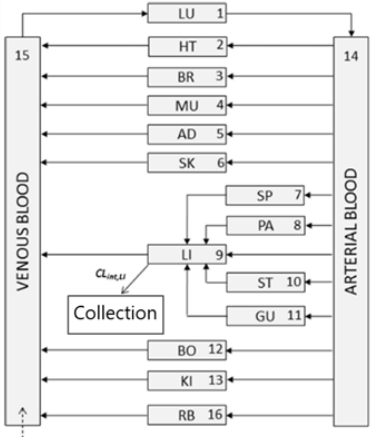
David Kolb's experiential learning cycle has much of the characteristics of an iterative design process constructed around experimentation, experience, engagement and learning. Based on Kolb, D. A. (1984). *Experiential learning: Experience as the source of learning and development (Vol. 1)*. Englewood Cliffs, NJ: Prentice-Hall

Occam's razor and why it is relevant

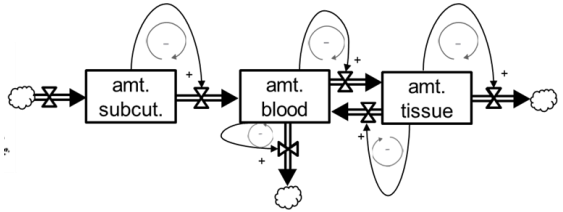


"We consider it a good principle to explain the phenomena by the simplest hypothesis possible."
- Ptolemy (circa 90-168 AD)

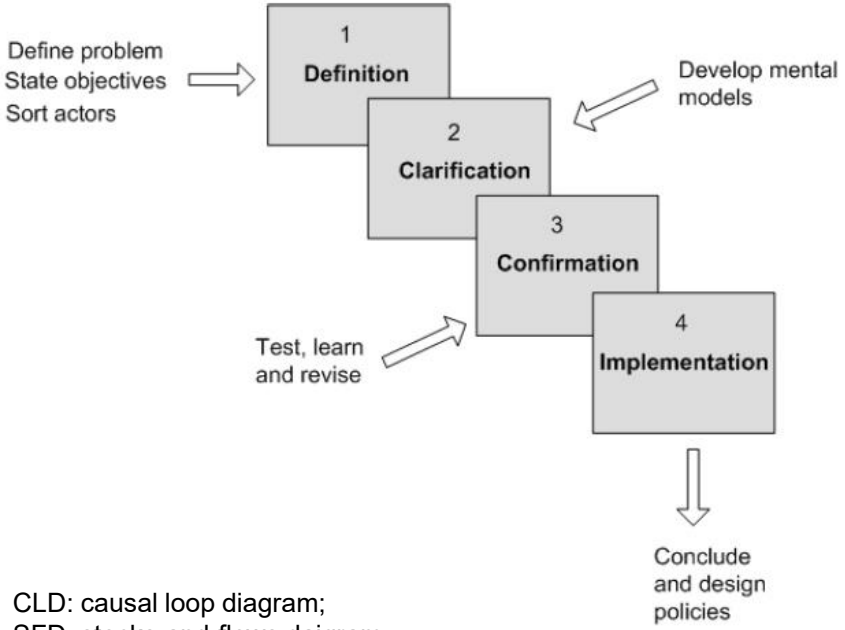
More complex models: More prone to bias and uncertainty. Loss of predictive power.



Simpler models: May not <u>explain</u> observed system behaviour at a sufficiently detailed/accurate level (bottom-up, mechanism).



Model implementation



CLD: causal loop diagram;
SFD: stocks-and-flows daigram;
SDTD: system dynamics tool diagram.
SBIC: system boundaries, inputs, *etc*.

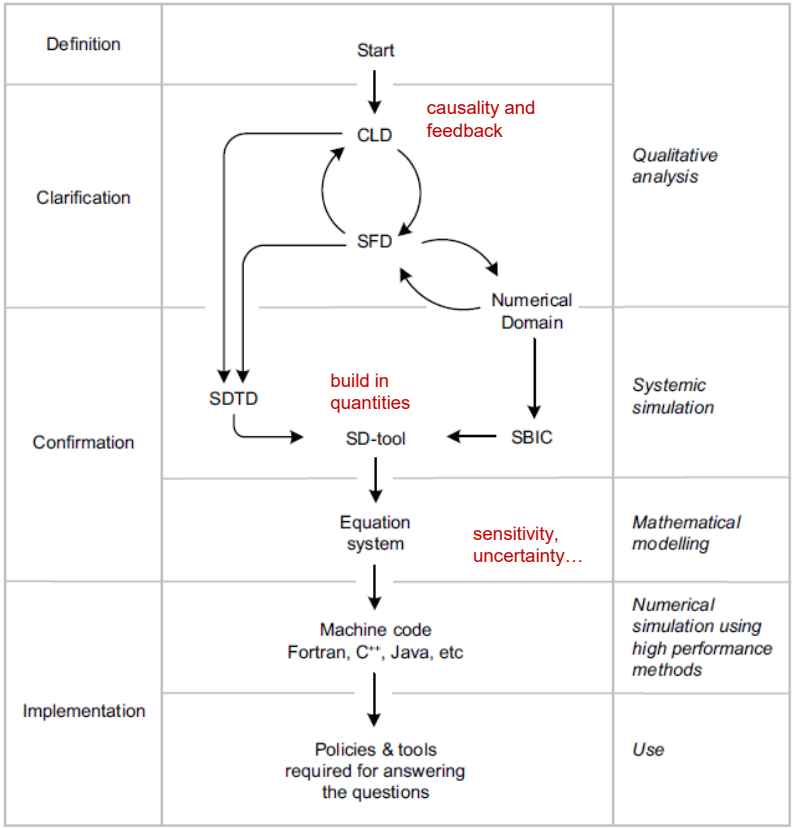
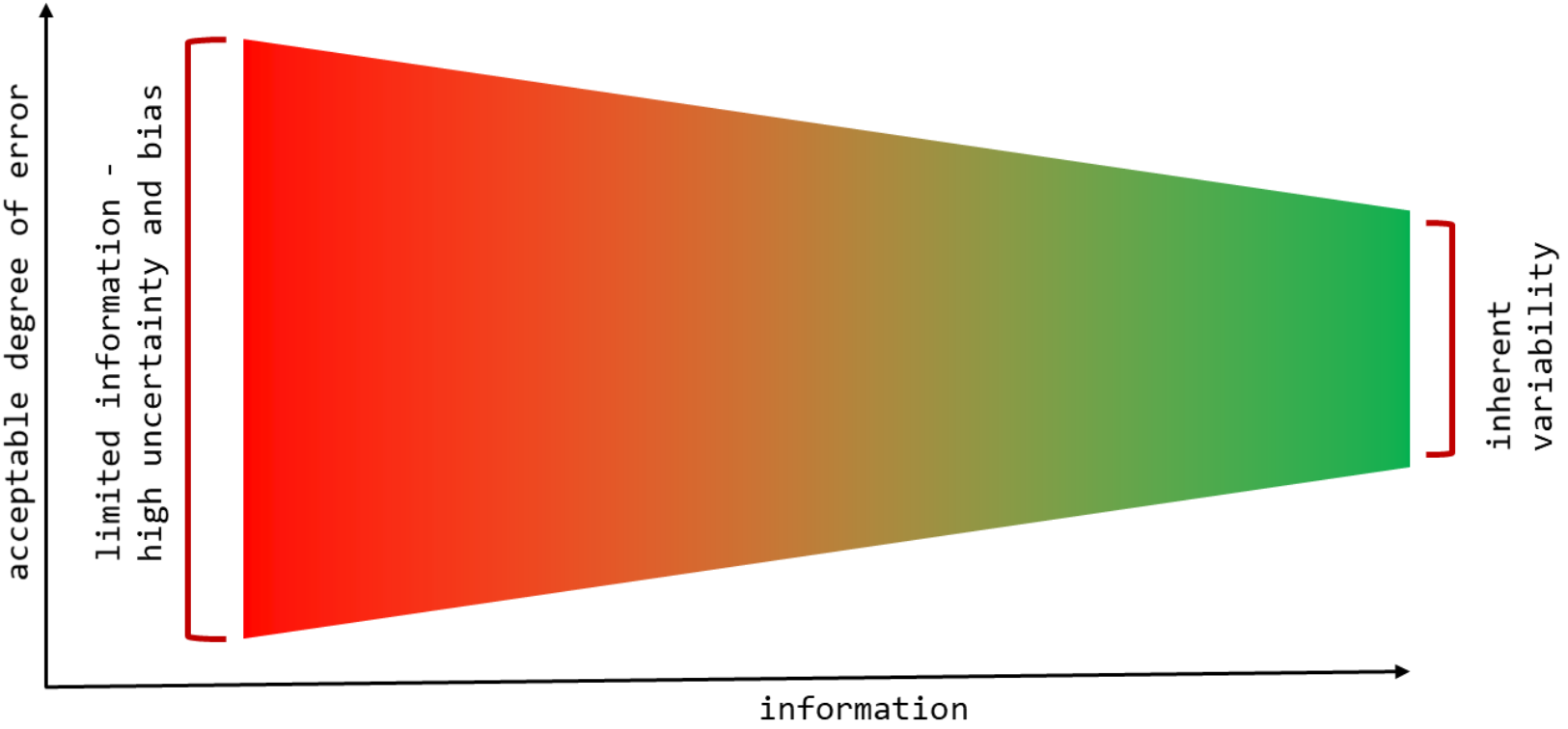


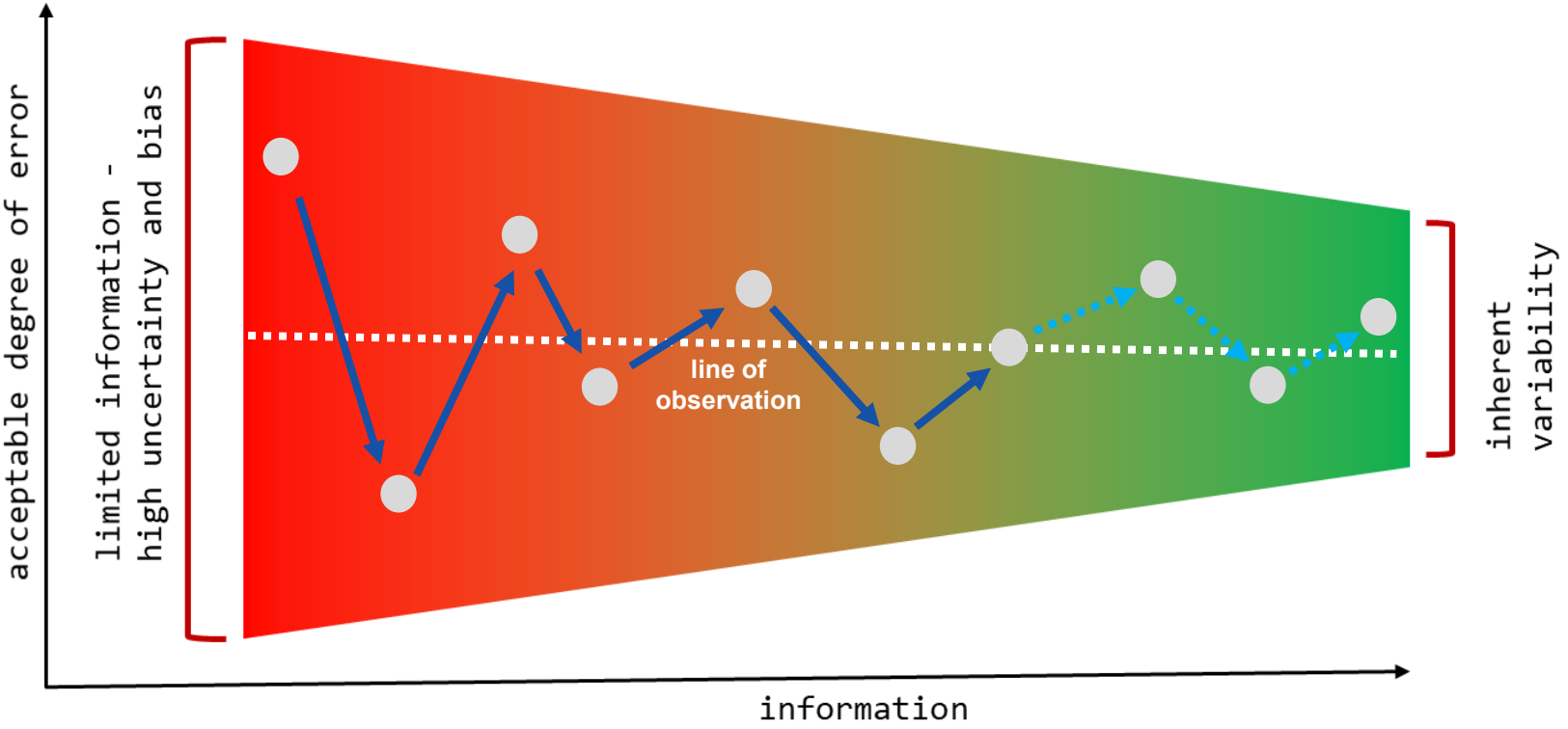
Figure 3.8. Summary of workflow in the modelling process, from qualitative analysis to building a numerical model, showing the relationship between the CLD, SFD, and SDTD. Each phase involves a different level of detail for analysing and answering the question.

Haraldsson and Svedrup 2020

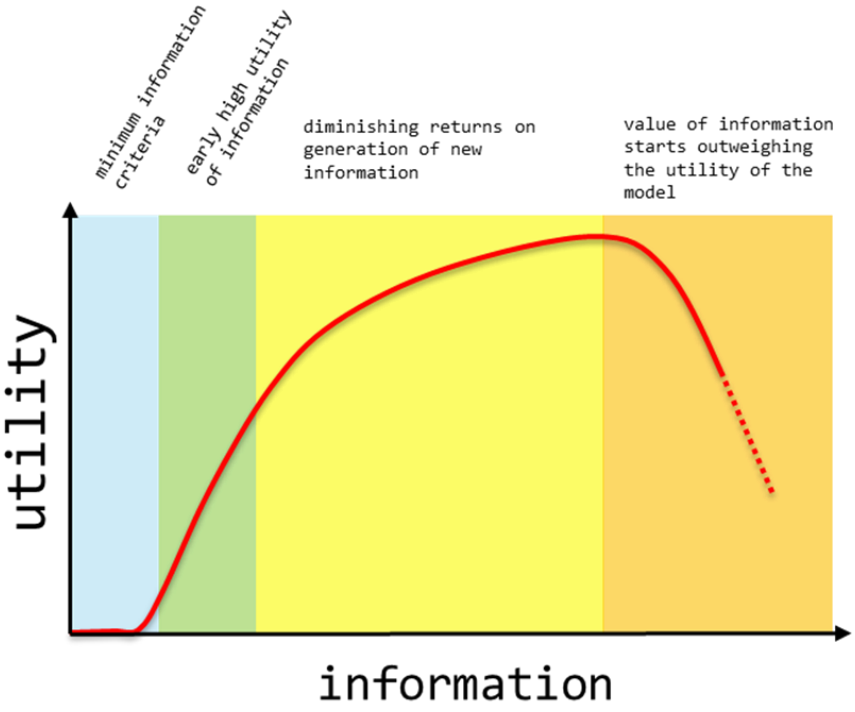
Data requirements, data availability, and criteria for success



Data requirements, data availability, and criteria for success



Utility of a model as a function of available data



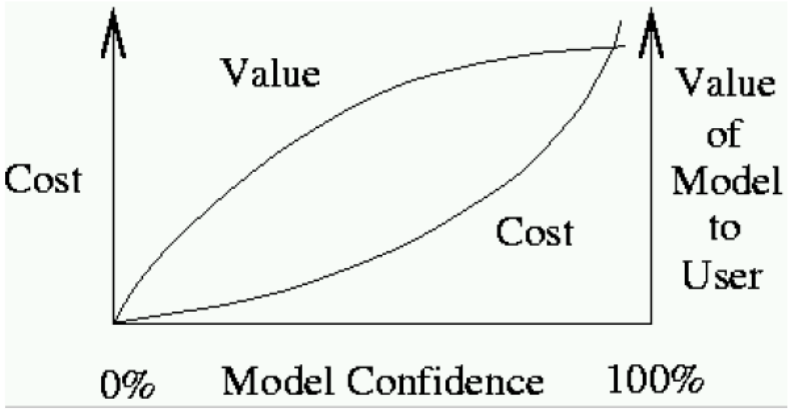
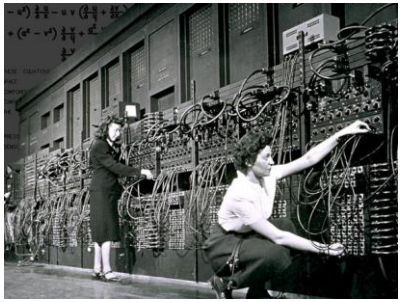


Figure 1: Model Confidence

Darwich 2021; Sargent 2010

Model verification



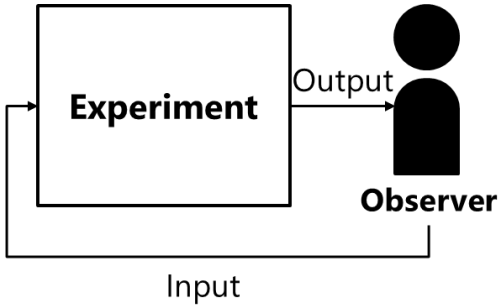
debugging



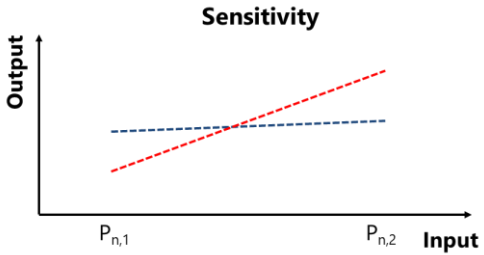
expert input

$$A = \sum_{i=1}^{n} V_i \cdot \frac{dC_i}{dt}$$

mass balance



uncertainty analysis



sensitivity analysis

- Can be defined as “ensuring that the computer program of the computerized model and its implementation are correct” (Schlesinger *et al*. 1979; Sargent 2010).
- “Model Verification is substantiating that the model is transformed from one form into another, as intended, with sufficient accuracy. Model verification deals with building the model right. The accuracy of transforming a problem formulation into a model specification or the accuracy of converting a model representation from a micro flowchart form into an executable computer program is evaluated in model verification” (Balci 1998).

Debugging

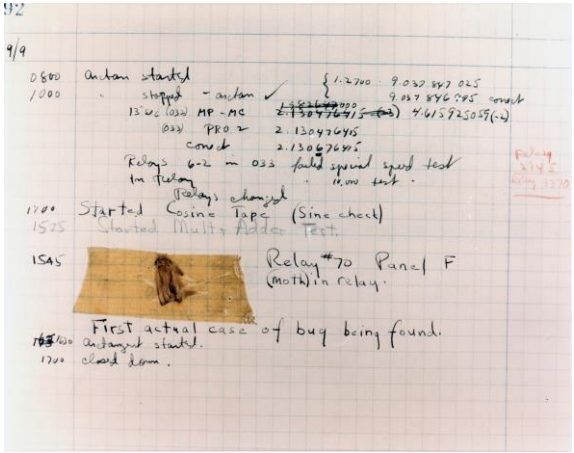
```
dA(n)=a(n-1)*k_(n-1->n) - a(n)*(k_(n->n+1)

dA(n+1)=a(n)*k_(n-1) - a(n+1)*k_(n+1->n+2)
```





Grace Hopper, Pioneer in programming



Smithsonian

Expert & stakeholder input: participatory methods



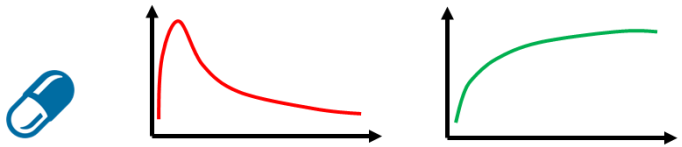


Used throughout:

- **Model development**
- **Verification**
- **Validation**
- **Application**

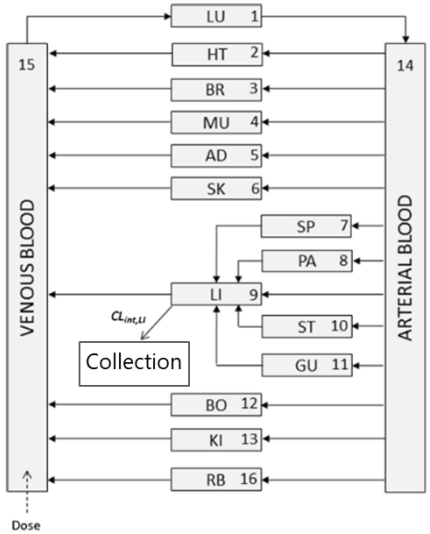
Hill *et al*. 2022

model verification: mass balance



Input=Accumulation + Output

Closed systems

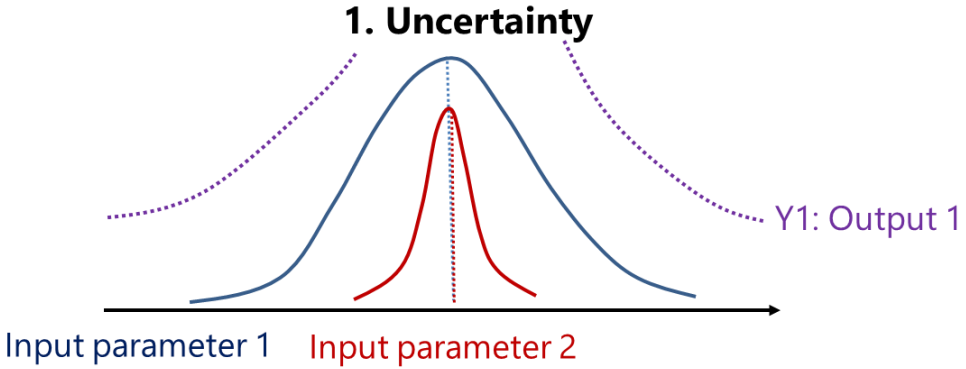


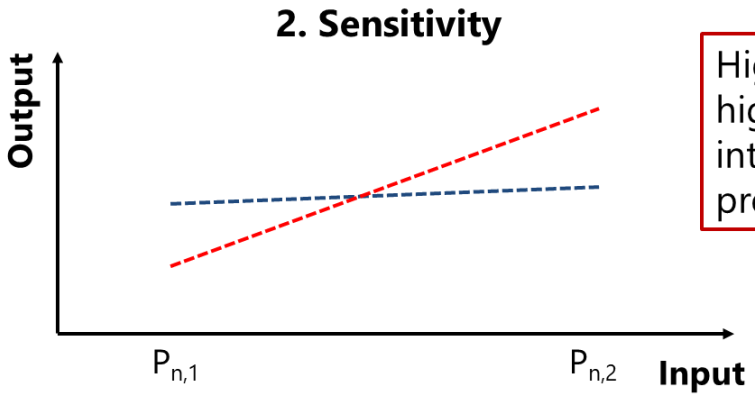
collection compartments, sinks.

$$A = \sum_{i=1}^{n} V_i \cdot \frac{dC_i}{dt}$$

Conservation of mass:
Matter cannot be created spontaneously or disappear.

What does uncertainty and sensitivity say about the reliability of a model?

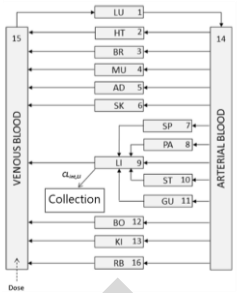




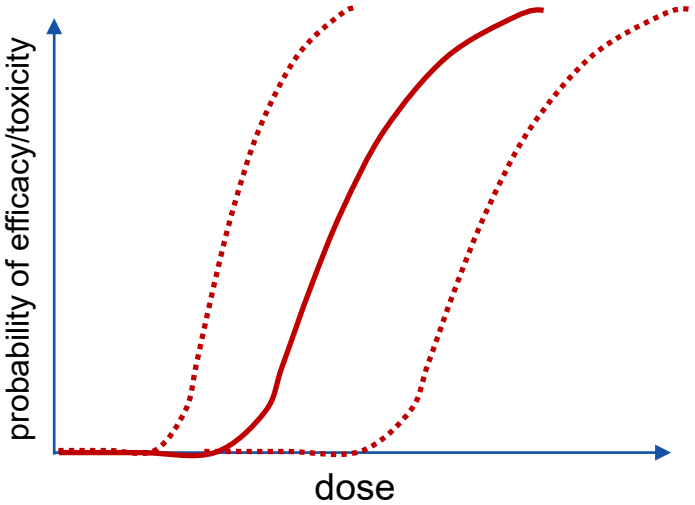
High uncertainty and high sensitivity translates into less reliability in model predictions.

An example on how we use uncertainty in practice

systems model based on preclinical information



simulation



what is the chance of a given outcome given a certain dose?

first dose in human (accounting for potential unknown errors)

inputs with variability (including uncertainty) , e.g.:
$i_j = \theta_j + \text{randn}(0,\eta_j)$

An example on how we use sensitivity analysis in practice

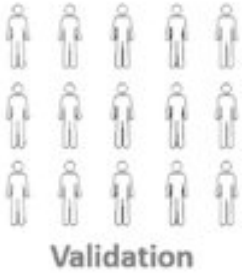
Model validation



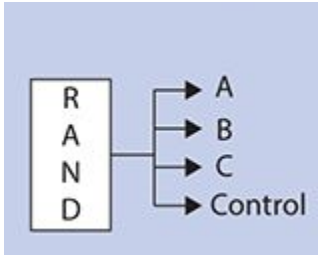
face validity



internal validation



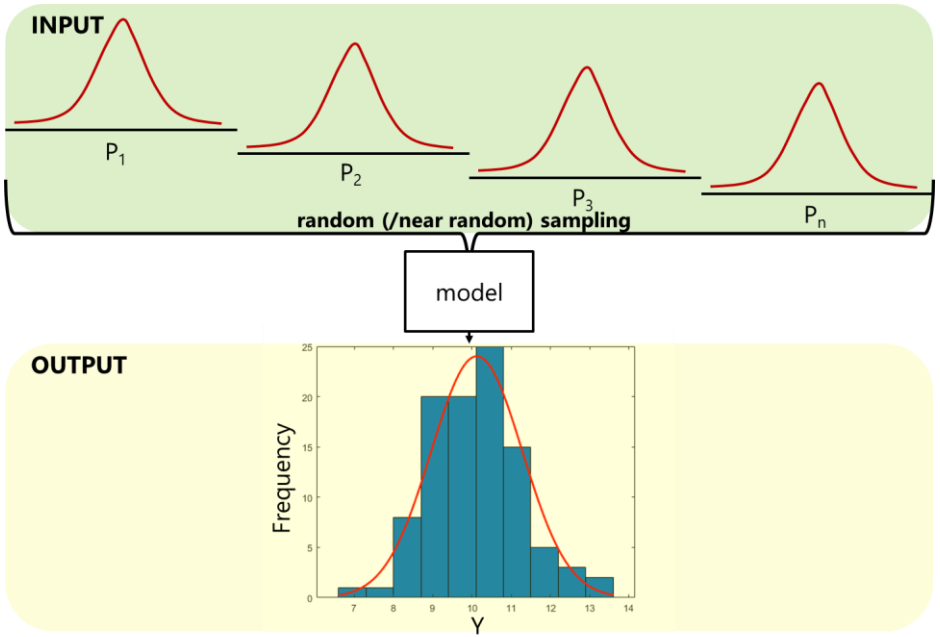
external validation

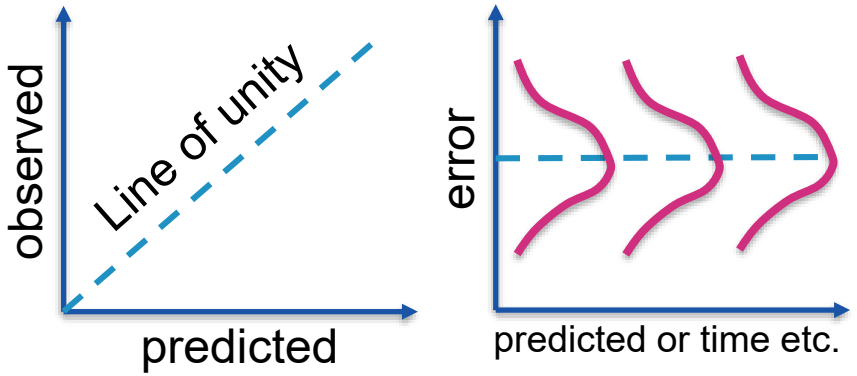


validation through perturbation/(extreme conditions)

- "Substantiation that a computerized model within its domain of applicability possesses a satisfactory range of accuracy consistent with the intended application of the model" (Schlesinger *et al*. 1979; Sargent 2010).
- "Model Validation is substantiating that the model, within its domain of applicability, behaves with satisfactory accuracy consistent with the M&S objectives. Model validation deals with building the right ["right"] model" (Balci 1998).

Model validation





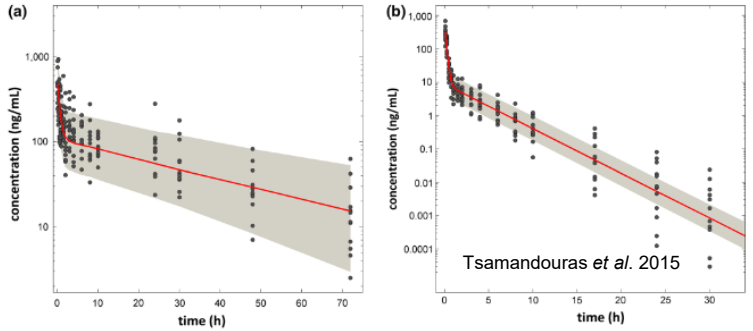
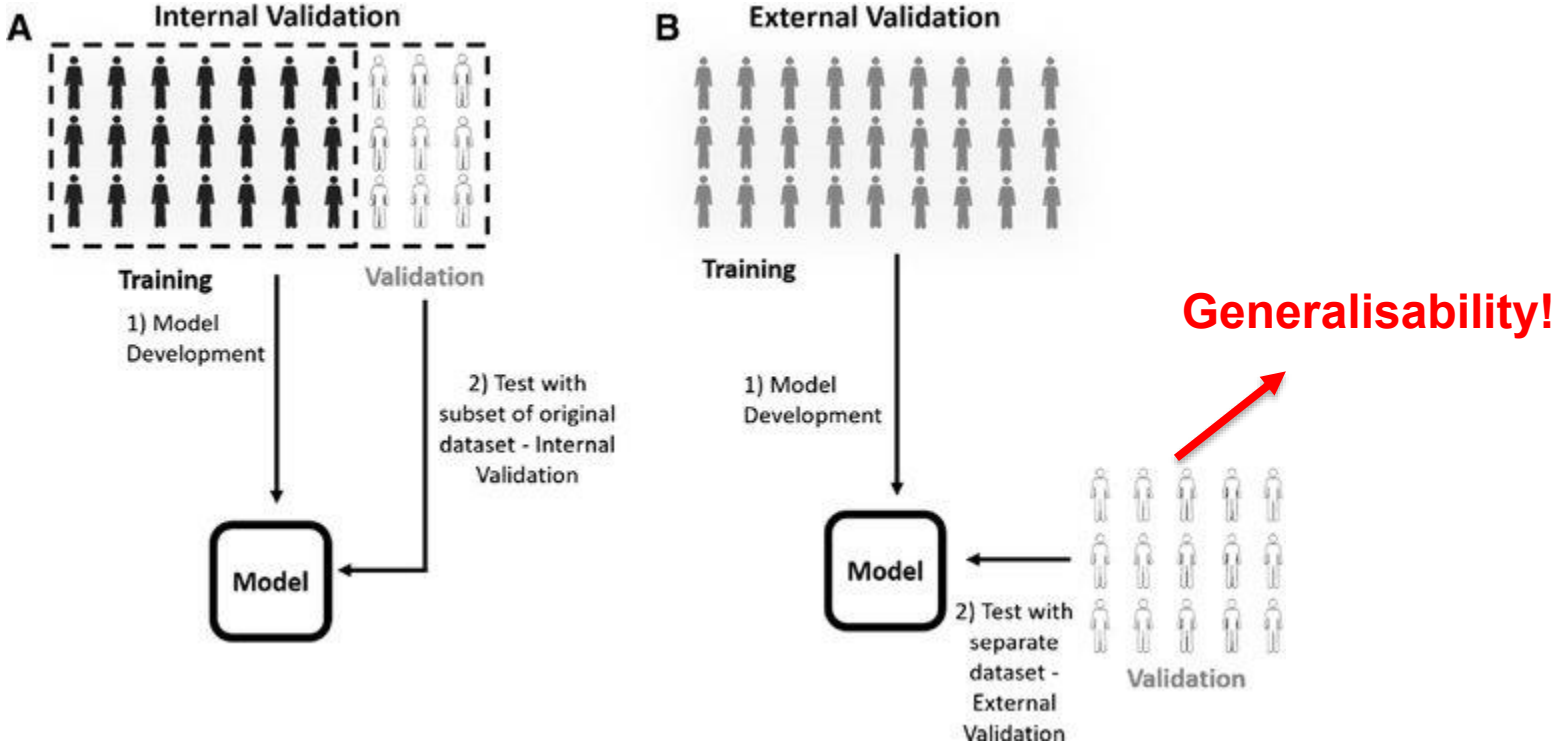


Fig. 13 Investigation of the ability of a diazepam population PBPK model with biased/misplaced stochastic level to describe fitted data (**a**) and perform extrapolation (**b**). In both (**a**) and (**b**), *grey area* and *red line* represent respectively 90 % population prediction interval and median obtained from model simulations. (**a**) Presented is a VPC corresponding to the fit of one of the simulated datasets (see main text). Specifically, the plotted diazepam plasma data have been simulated with a population PBPK model that assumes a 30 % CV in organ blood flows and volumes. However, the fitted model omits this system parameter variability and instead, in order to compensate and capture the variability in the data, inflates IIV_{CLint} by 55 % and ResV by 34 % from their real values (values used to generate the data). (**b**) Diazepam plasma concentration data have been generated again with a population PBPK model that assumes a 30 % CV in organ blood flows and volumes but this time after extrapolation to the high E_H case. The plotted median and population prediction intervals refer to the diazepam model with biased stochastic structure as described in (**a**), after extrapolation to the high E_H case (Color figure online)

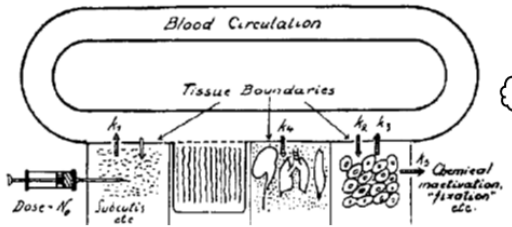
Internal and external validation

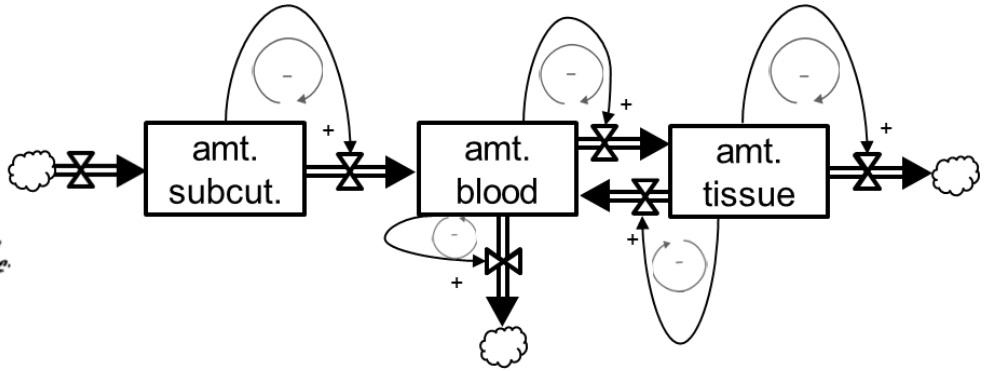


Validation through model perturbation or extreme conditions

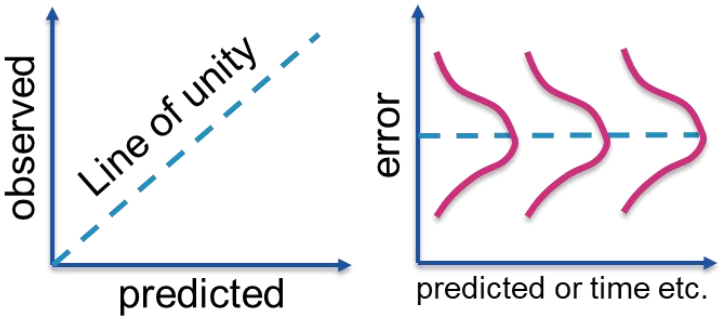
- How does the model perform against observed data when we push input parameters to their boundaries?
- How does the model perform under different scenarios/conditions?

Examples from systems pharmacology:
What happens at very low, or very high, doses?
What happens when we inhibit drug elimination?
What happens if we change rates to represent a rat, dog, human?





Prediction



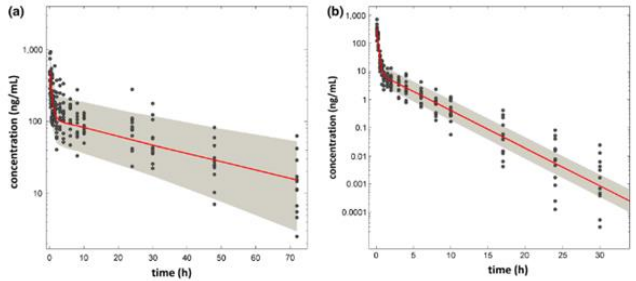
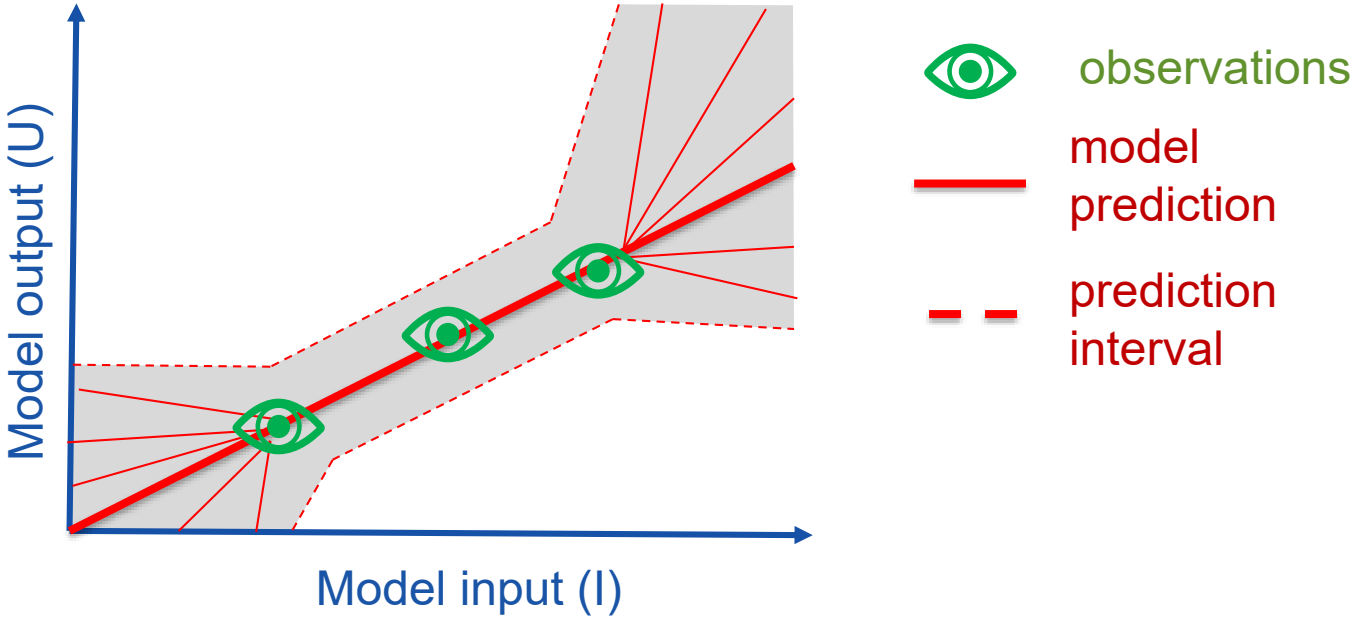
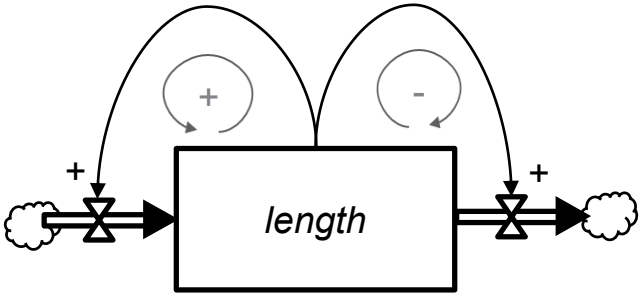


Fig. 13 Investigation of the ability of a diazepam population PBPK model with biased/misplaced stochastic level to describe fitted data (**a**) and perform extrapolation (**b**). In both (**a**) and (**b**), *grey area* and *red line* represent respectively 90 % population prediction interval and median obtained from model simulations. (**a**) Presented is a VPC corresponding to the fit of one of the simulated datasets (see main text). Specifically, the plotted diazepam plasma data have been simulated with a population PBPK model that assumes a 30 % CV in organ blood flows and volumes. However, the fitted model omits this system parameter variability and instead, in order to compensate and capture the variability in the data, inflates $\text{IIV}_{\text{CLint}}$ by 55 % and ResV by 34 % from their real values (values used to generate the data). (**b**) Diazepam plasma concentration data have been generated again with a population PBPK model that assumes a 30 % CV in organ blood flows and volumes but this time after extrapolation to the high E_H case. The plotted median and population prediction intervals refer to the diazepam model with biased stochastic structure as described in (**a**), after extrapolation to the high E_H case (Color figure online)

Prediction: interpolation vs. extrapolation



Prediction: interpolation vs. extrapolation



$$\frac{dL_{animal}}{dt} = L_{animal}(t)^{\alpha} \cdot p - L_{animal}(t) \cdot q$$

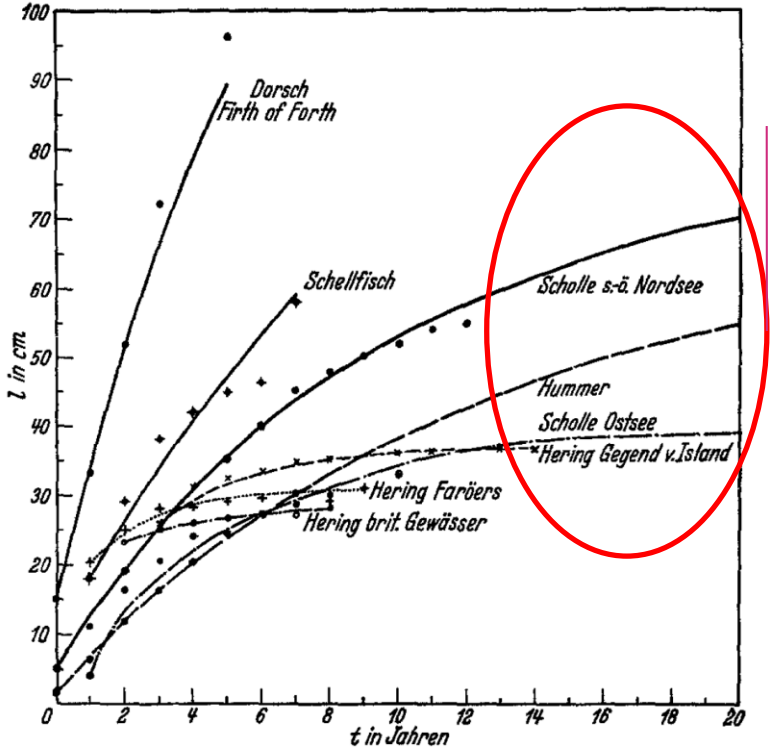


Abb. 4. Längenwachstum von Meerestieren, nach Angaben von PÜTTER (1920), die berechneten Kurven ausgezogen, mit Bezeichnung der empirischen Bestimmungen. (Einige mit anderen zusammenfallende Bestimmungspunkte wurden weggelassen.)

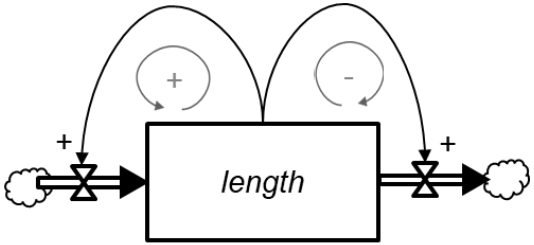
How confident can we be regarding what is going on outside our observed data points?

Von Bertalanffy 1934

Prediction: interpolation vs. extrapolation

Biological growth

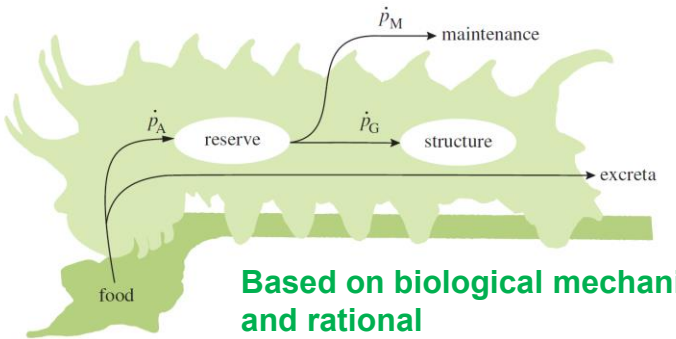
Empirical, data-driven models
+ interpolation



$$\frac{dL_{animal}}{dt} = L_{animal}(t)^{\alpha} \cdot p - L_{animal}(t) \cdot q$$

Based on data

Mechanistic models
+ extrapolation

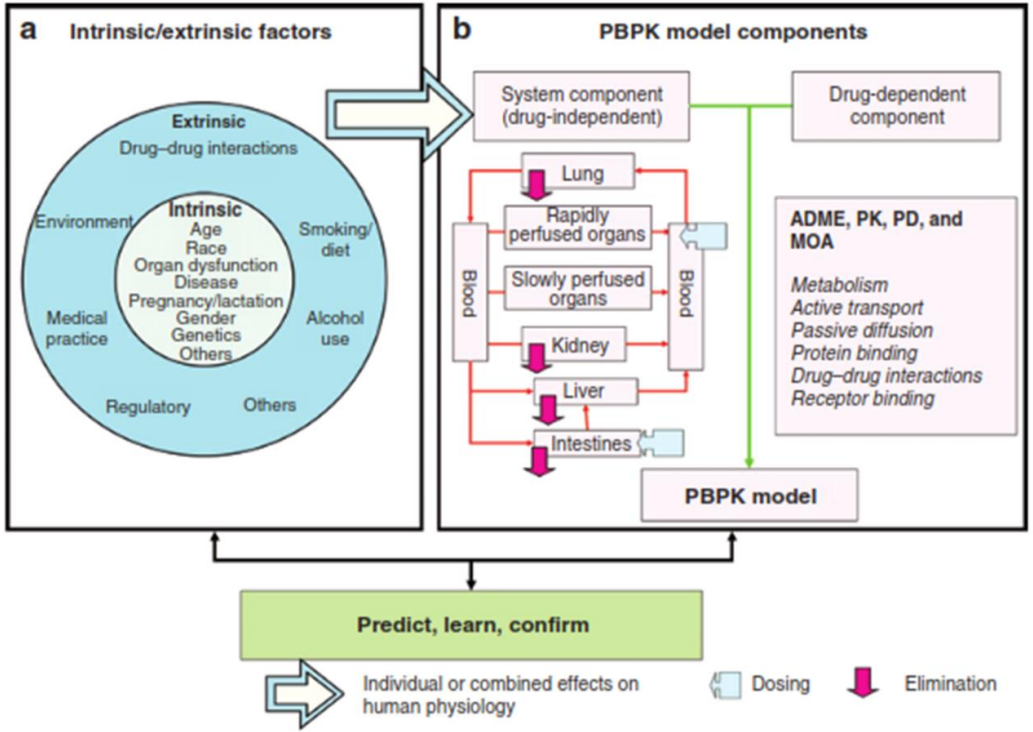


Based on biological mechanisms and rational

Figure 2. A simplified schematic of a standard DEB model without maturation or reproduction. DEB theory uniquely partitions organism biomass into reserve and structure compartments. For the sake of simplicity, we do not consider allocation to maturation or reproduction, which would usually be treated as an extra branch before allocation to growth and maintenance. (Online version in colour.)

Von Bertalanffy 1934; Maino 2015

Predict-learn-confirm



The degree of complexity of the PBPK model can vary according to the need.

Optimisation – considerations:

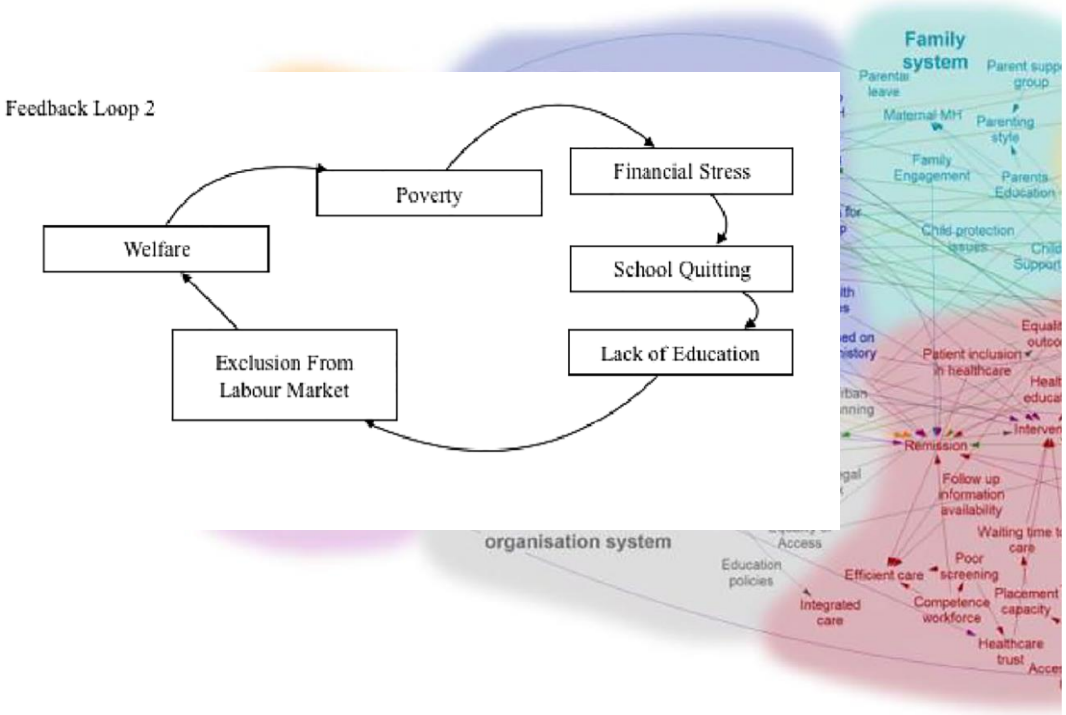
- Identifiability and sensitivity
- Choice of objective function
- Minimisation method
- Error model used
- Mechanistic plausibility

Zhao *et al*. 2011 – FDA; IPS; WHO 2010

Table 10. Summary of best-practice statements with highest importance and high agreement

Stage	Rule-like statements (prescriptive)
In exemplary *problem identification and definition,* you should . . .	Speak with and listen carefully and reflectively to problem owners (clients) to identify and understand the problem Explicitly clarify and state the purpose of the modeling effort (e.g., strategy development, policy analysis, theory building, education, training) Identify the reference modes of central processes to be studied for the purpose of clarifying expectations of future behavior Ask why the current behavior of key variables generated is the way it presently is, and what is causing it Formulate a dynamic hypothesis
In exemplary *system conceptualization*, you should . . .	Approach the conceptualization process creatively and from different angles, avoiding a rigid separation of the identification and conceptualization stages Generate a dialogue with the problem's owners that focuses on their mental models and on dynamic hypotheses Identify the critical accumulations (stocks) that describe the system, making certain that their names are nouns — not verbs or action phrases
In exemplary *model formulation*, you should . . .	Develop the structure through a series of simple to more comprehensive models, adding detail as needed to improve realism and show policy impacts Formulate equations that make sense, carefully supporting variables and parameters with data or experience and making certain that all have real-life meanings Make sure that the model always exhibits dimensional consistency
In exemplary *model testing and evaluation*, you should . . .	Compare simulated behavior patterns with real behavior (data) using statistical measures of pattern fit, not point-by-point fit
In exemplary *model use, implementation, and dissemination*, you should . . .	Make sure that the entire modeling process revolves around the problems of concern of the audience (problem owner, client)
In exemplary *design of learning strategy/infrastructure*, you should . . .	Use simplified causal-loop diagrams to tell system stories in a variety of ways, rather than relying on the model to tell its own story

System dynamics modelling and simulation to support decision-making: mental health



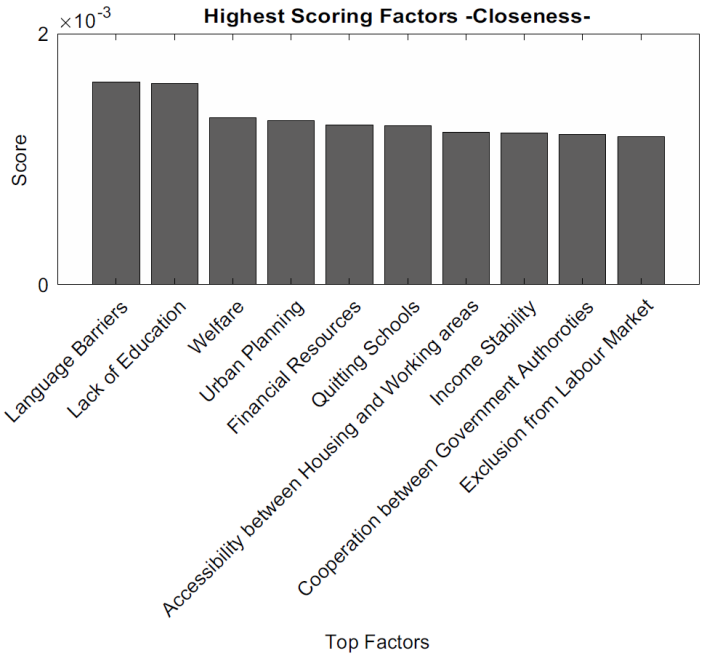
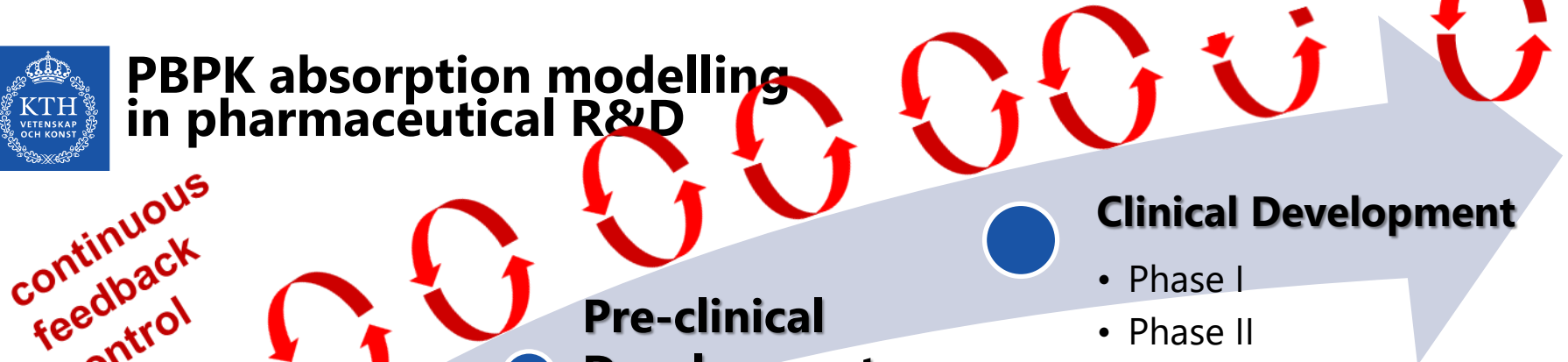


Fig. 7 Top closeness scoring factors

(Moustaid *et al*. 2020)

PBPK absorption modelling in pharmaceutical R&D



Drug Discovery

- Lead Optimisation
  - Early stage prediction of absorption
  - Explore limitations using PSA
- Explore formulations
- Development of formulations for toxicology studies
- Predict oral pharmacokinetics in human
- Define clinical formulation strategy

Pre-clinical Development

- Clinical Lead Selection
- Entry Into Human

Clinical Development

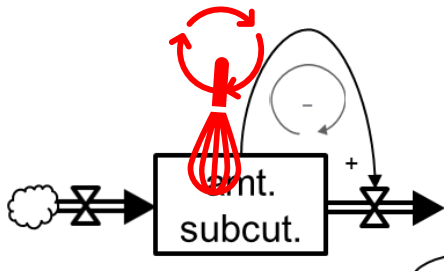
- Phase I
- Phase II
- Phase III
  - Predict food effects
  - Design extended release formulations
  - Develop IVIVCs

- Increasing amount of data
- Increasing model complexity
- Increasing level of validation

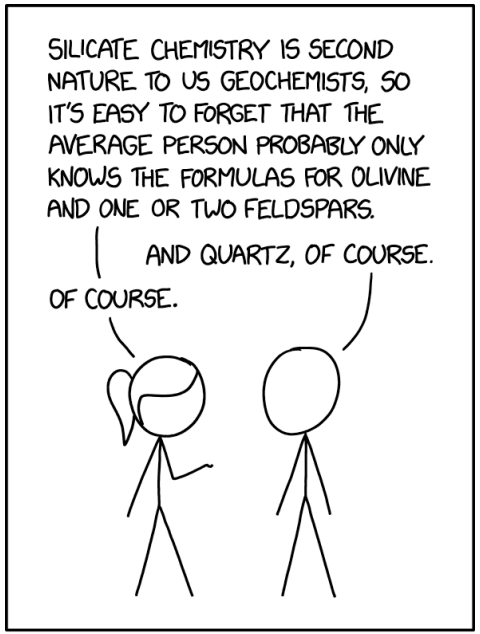
Adapted from Parrott and Lavé (2008)

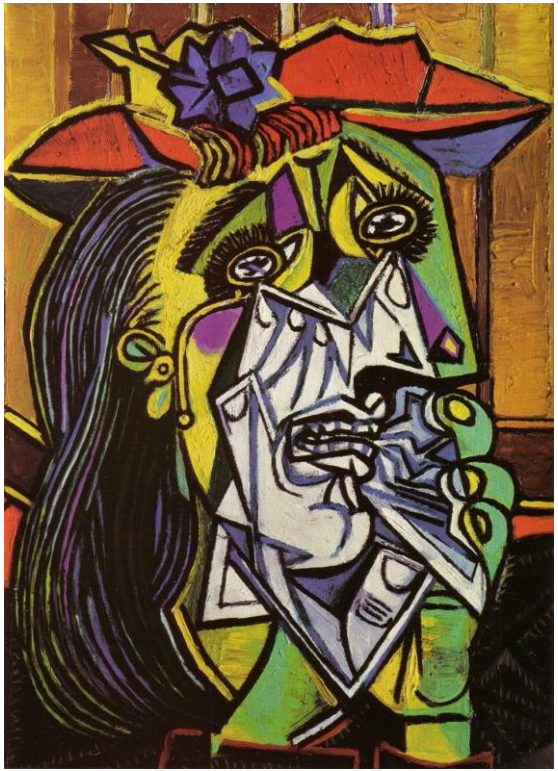
Advantages and limits of system dynamics simulations

- System dynamic models may be considered abstract with a high degree of aggregation (loss of resolution).
- These models are therefore suitable for studying problems at a higher level of abstraction, for example: population level, concentrations, volumes, amounts – over time.
- States (stocks) are assumed to be "well-stirred" and homogenous.
  - Does not consider individual entities inside a given state.
- Processes are defined as rates or flows.
  - Does not consider discrete events (such as single cell death, patient transitions in disease/healthcare logistics models, *etc.*).



Different levels of abstraction and resolution





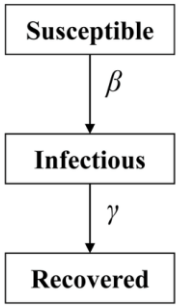


System dynamics → Discrete event simulation → Agent-based modelling

process stocks, flows and delays.

series of discrete events – Focus on process flow of entitites, queues and servers.

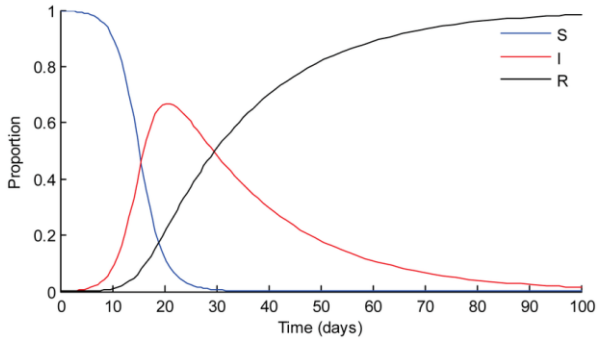
focus on individual entities, their attributes, behaviour and interactions.

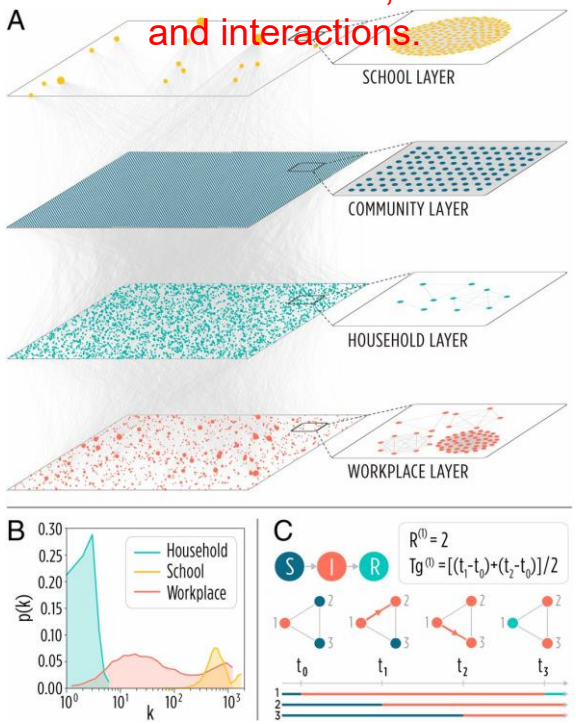


$$\frac{dS}{dt} = -\beta SI$$

$$\frac{dI}{dt} = \beta SI - \gamma I$$

$$\frac{dR}{dt} = \gamma I$$





System dynamics → Discrete event simulation → Agent-based modelling

SD+DES - Highly complementary to capture processes mechanstically and at different levels of resolution

top-down(?)
deterministic(?)

middle-out(?)
stochastic(?)

bottom-up(?)
stochastic(?)

Agent-theorem for system dynamics: every well-defined SD problem could be defined as an ABM (Macal *et al*. 2010).

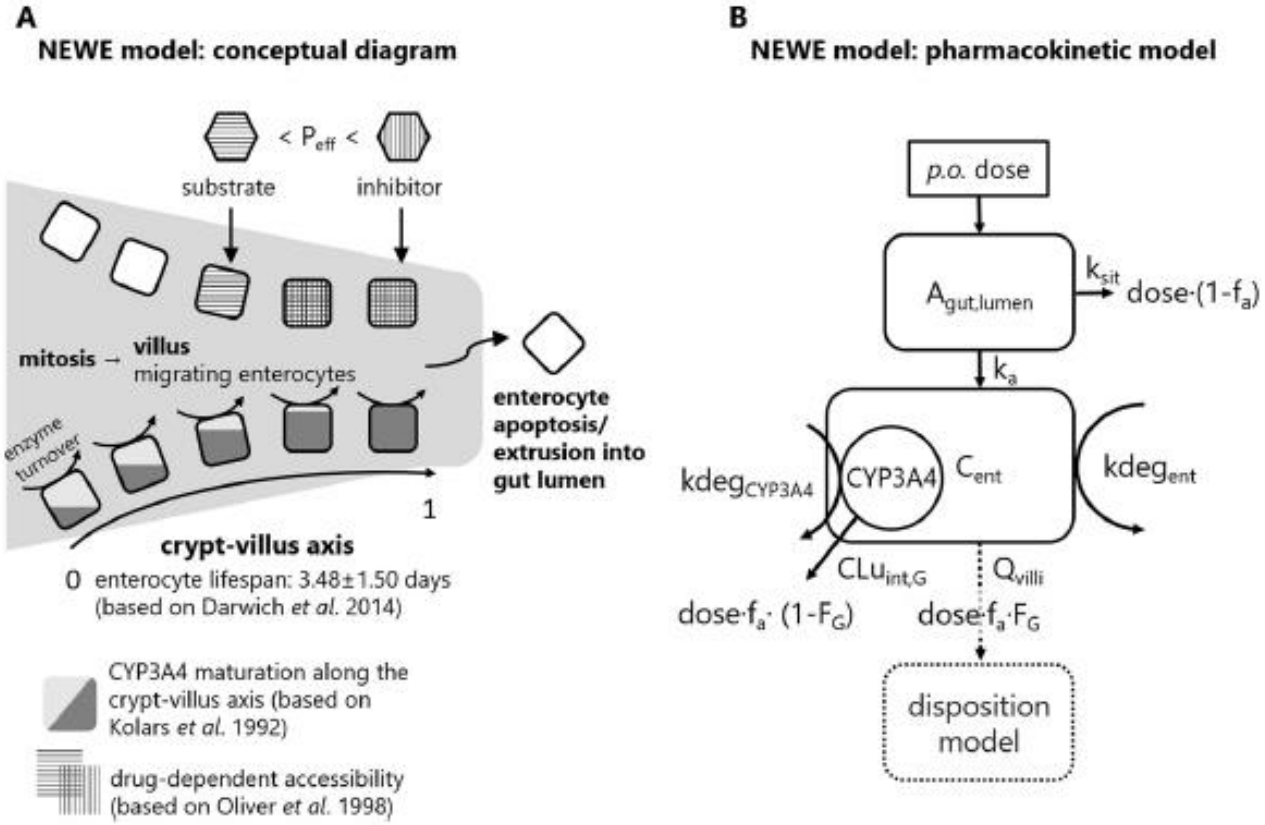
passive entities

active entities with unique behaviour and interactions.

In systems modelling there is value to all of these. And, all combinations are viable. This depends on the problem.

high predictive value

high explainable value.

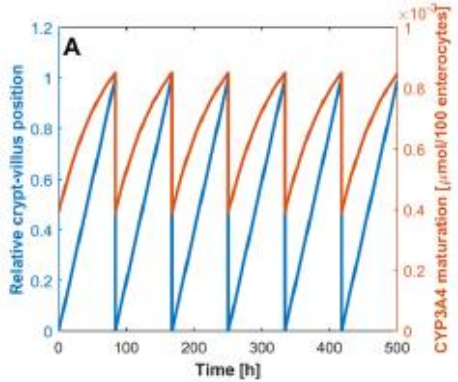


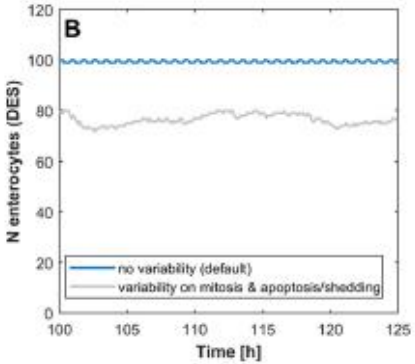
Example:
SD-DES combo

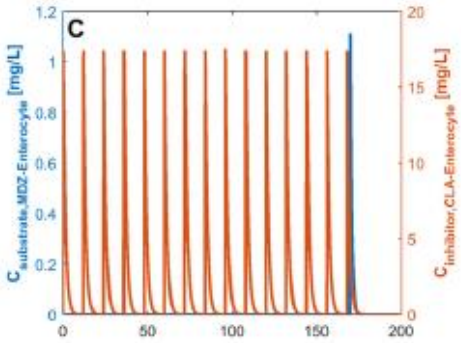
The nested enzyme-within-enterocyte (NEWE) turnover model for predicting dynamic drug and disease effects on the gut wall

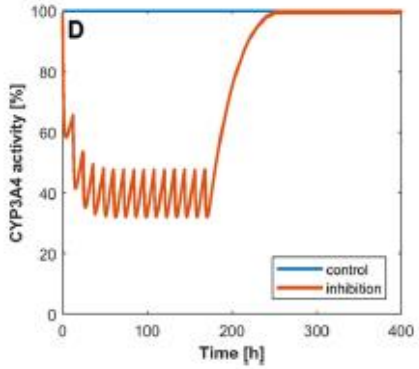
Adam S. Darwich[a,*], Howard J. Burt[b], Amin Rostami-Hodjegan[a,b]

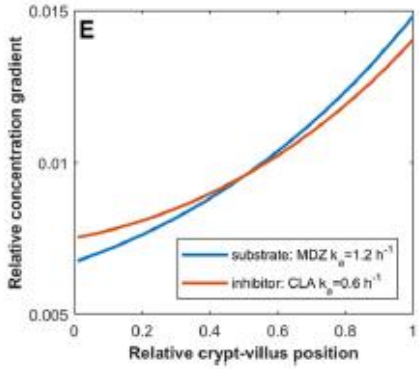
[a] Centre for Applied Pharmacokinetic Research, School of Health Sciences, The University of Manchester, Manchester, United Kingdom
[b] Certara UK Ltd., Sheffield, United Kingdom











The nested enzyme-within-enterocyte (NEWE) turnover model for predicting dynamic drug and disease effects on the gut wall

Adam S. Darwich[a,*], Howard J. Burt[b], Amin Rostami-Hodjegan[a,b]

[a] Centre for Applied Pharmacokinetic Research, School of Health Sciences, The University of Manchester, Manchester, United Kingdom
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SD-DES combo

Summary

- Here we discuss some of the important aspects in model development and examples of good practices.
- The general approach is valid across several modelling techniques and disciplines.
- However, there is no prescriptive framework for what a ‘best approach’ should look like. This will depend on what methodology is used, discipline specific practices, stakeholder engagement, regulation, and more.
- **Do not let dogma be an obstacle to progress and creativity!**



Jackson Pollock



KTH ROYAL INSTITUTE
OF TECHNOLOGY

contact: darwich@kth.se

